Principles of Anatomy and Physiology

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Sixteenth Edition

Chapter 2

The Chemical Level of Organization

Introduction

The purpose of this chapter is to:

- Introduce the language and fundamental concepts of chemistry
- Discuss how matter is organized
- Discuss how chemical bonds form and how chemical reactions occur
- Compare and contrast organic and inorganic compounds

How Matter Is Organized

Basic Principles of Chemistry

- Chemistry is the science of structure and interactions of matter
- Matter is anything that has mass and takes up space
- Mass is the amount of matter in an object, whereas weight is the force of gravity acting on a mass

Chemical Elements

- Matter exists in 3 forms:
  - Solid
  - Liquid
  - Gas
- All forms of matter are composed of chemical elements

Elements

- Elements are given chemical symbols such as:
  - O = oxygen
  - C = carbon
  - H = hydrogen
  - N = nitrogen

- These elements make up the majority of our bodies

Main Chemical Elements in the Body

Chemical Element (Symbol)	% of Total Body Mass	Significance
Major Elements	(about 96)	
Oxygen (0)	65.0	Part of water and many organic (carbon-containing) molecules; used to generate ATP, a molecule used by cells to temporarily store chemical energy.
Carbon (C)	18.5	Forms backbone chains and rings of all organic molecules: carbohydrates, lipids (fats), proteins, and nucleic acids (0NA and RNA).
Hydrogen (H)	9.5	Constituent of water and most organic molecules; ionized form (H^+) makes body fluids more acidic.

Main Chemical Elements in the Body

Chemical Element (Symbol)	% of Total Body Mass	Significance
Nitrogen (N)	3.2	Component of all proteins and nucleic acids.
Lesser Elements	(about 3.6)	
Calcium (Ca)	1.5	Contributes to hardness of bones and teeth; ionized form (Ca^{2+}) needed for blood clotting, release of some hormones, contraction of muscle, and many other processes.
Phosphorus (P)	1.0	Component of nucleic acids and ATP; required for normal bone and tooth structure.
Potassium (K)	0.35	Ionized form (K^{+}) is the most plentiful cation (positively charged particle) in intracellular fluid; needed to generate action potentials.

Main Chemical Elements in the Body

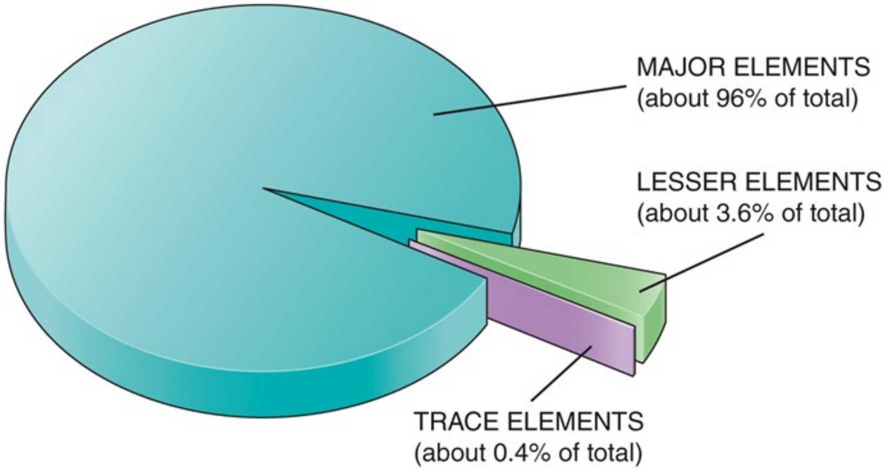
Chemical Element (Symbol)	% of Total Body Mass	Significance
Sulfur (S)	0.25	Component of some vitamins and many proteins.
Sodium (Na)	0.2	Ionized form (Na^+) is the most plentiful cation in extracellular fluid; essential for maintaining water balance; needed to generate action potentials.
Chlorine (Cl)	0.2	Ionized form (Cl^-) is the most plentiful anion (negatively charged particle) in extracellular fluid; essential for maintaining water balance.
Magnesium (Mg)	0.1	Ionized form (Mg^{2+}) needed for action of many enzymes (molecules that increase the rate of chemical reactions in organisms).

Main Chemical Elements in the Body

Chemical Element (Symbol)	% of Total Body Mass	Significance
Iron (Fe)	0.005	Ionized forms (Fe^{2+} and Fe^{3+}) are part of hemoglobin (oxygen-carrying protein in red blood cells) and some enzymes.

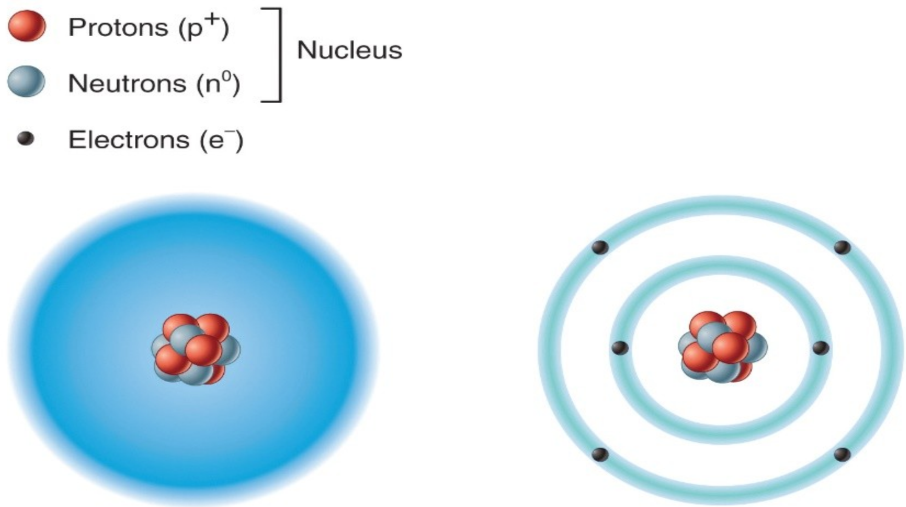
Main Chemical Elements in the Body

Chemical Element (Symbol)	% of Total Body Mass	Significance
Trace Elements	(about 0.4)	Aluminum (Al), boron (B), chromium (Cr), cobalt (Co), copper (Cu), fluorine (F), iodine (I), manganese (Mn), molybdenum (Mo), selenium (Se), silicon (Si), tin (Sn), vanadium (V), and zinc (Zn).



Atoms

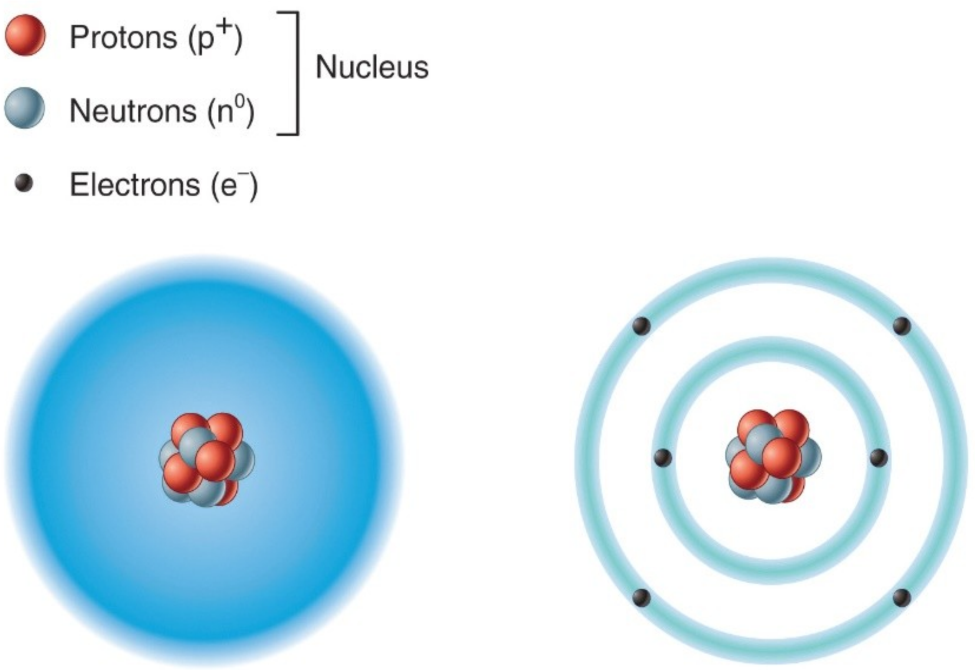
- Chemical elements are composed of units of matter of the same type called atoms
- Atoms are the smallest units of matter that retain the properties and characteristics of an element



(a) Electron cloud model (b) Electron shell model

Atomic Structure

- Central nucleus
    - Protons – positively charged
    - Neutrons – no charge
- Electrons
    - Negatively charged
    - Orbit around nucleus

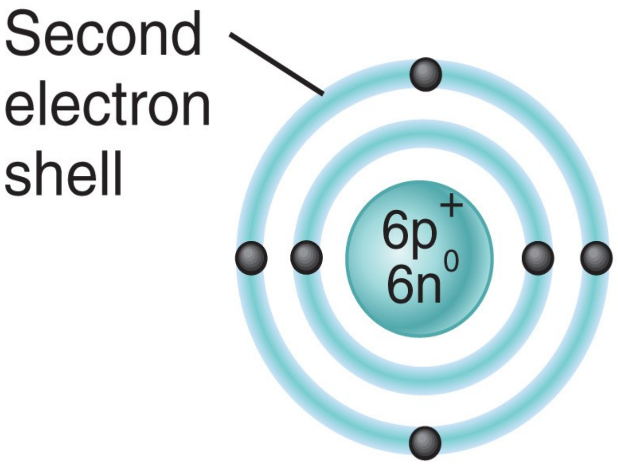


(a) Electron cloud model

(b) Electron shell model

Atomic Number and Mass Number

- Atomic number is the number of protons in the nucleus of an atom
- Mass number is the number of protons and neutrons in an atom
  - Isotopes
    - Atoms that have the same atomic number but differ in their number of neutrons (have different atomic masses)



Carbon (C)
Atomic number = 6
Mass number = **12** or 13
Atomic mass = 12.01

Radioactive Isotopes

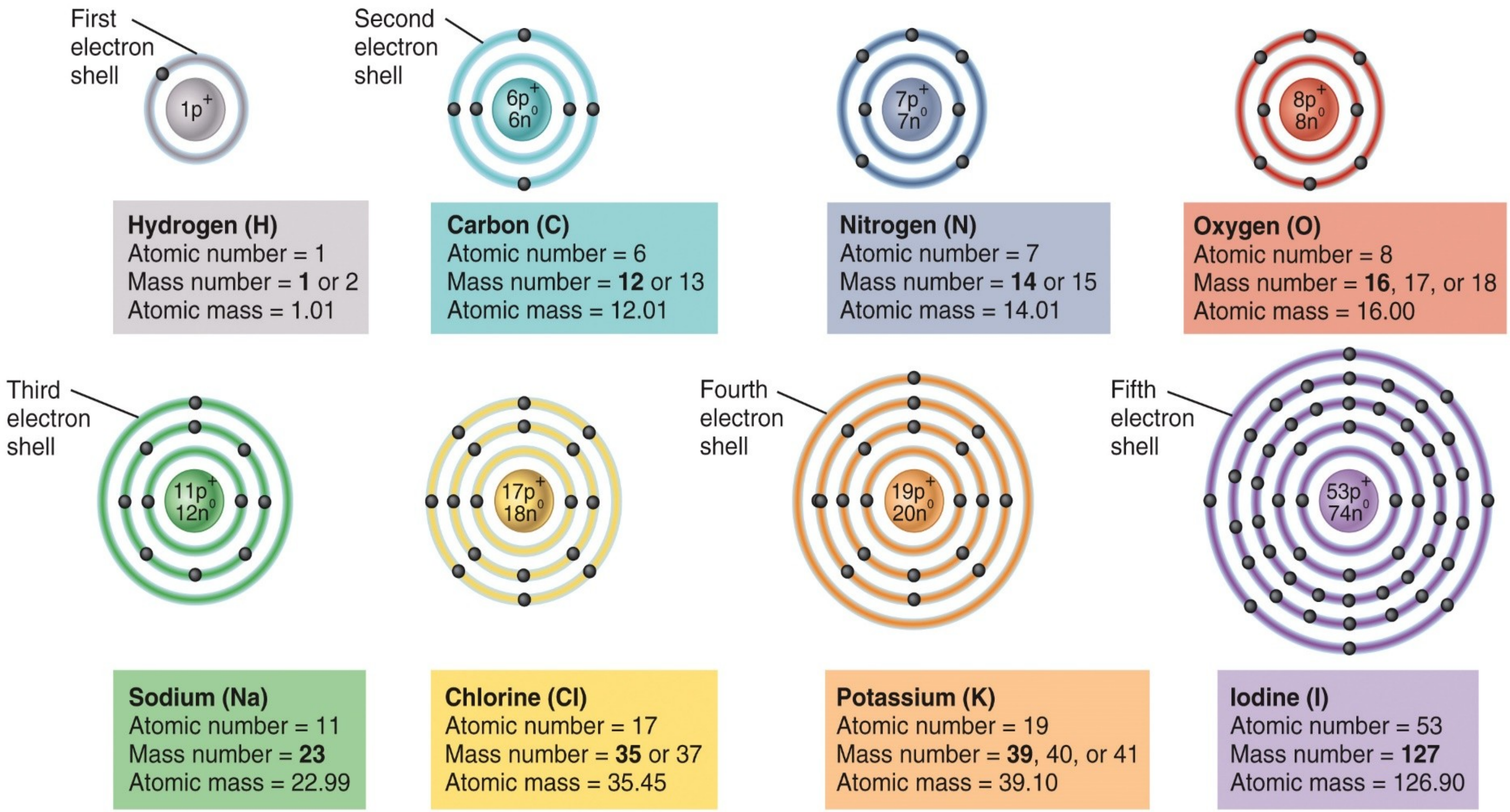
- Are unstable and break down/decay
  - Radioactive decay releases large amount of radiation energy
    - Depending on isotope, this radiation can be very damaging to cells
  - Decay is measured in half-lives
    - Amount of time it takes for ½ of the starting amount of the isotope to break down into a more stable form

Atomic Mass

- Atomic mass assumes the mass of a:
  - Neutron = 1.008 daltons
  - Proton = 1.007 daltons
  - Electron = 0.0005 daltons

- The atomic mass of an element is the average mass of all its naturally occurring isotopes

Atomic Structures



Atomic number = number of protons in an atom
Mass number = number of protons and neutrons in an atom (boldface indicates most common isotope)
Atomic mass = average mass of all stable atoms of a given element in daltons

Ions, Molecules, and Compounds

- Ion – an atom that has lost or gained an electron
  - Cation
    - Lost an electron
    - Positively charged
  - Anion
    - Gained an electron
    - Negatively charged
- Molecule – 2 or more atoms sharing electrons
- Compound – a substance that can be broken down into 2 or more different elements

Ions: Cations

Name	Symbol
Hydrogen ion	H^+
Sodium ion	Na^+
Potassium ion	K^+
Ammonium ion	NH_4^+
Magnesium ion	Mg^{2+}
Calcium ion	Ca^{2+}
Iron (II) ion	Fe^{2+}
Iron (III) ion	Fe^{3+}

Ions: Anions

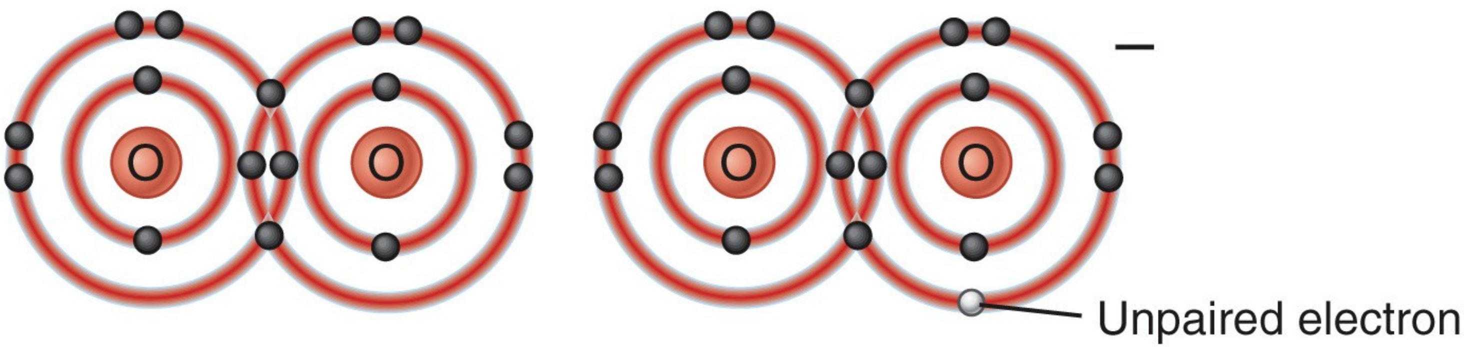
Name	Symbol
Fluoride ion	F^-
Chloride ion	Cl^-
Iodide ion	I^-
Hydroxide ion	OH^-
Bicarbonate ion	HCO_3^-
Oxide ion	O^{2-}
Sulfate ion	SO_4^{2-}
Phosphate ion	PO_4^{3-}

Orbitals, Shells, and Valence Electrons

- Electron orbitals
  - Area around the nucleus where an electron is most likely to be found
  - Fits a maximum of two electrons
- Energy shells
  - 1st (closest to nucleus) is composed of one orbital
  - 2nd and 3rd each composed to 4 orbitals
- Valence electrons
  - Electrons in the outermost shell
  - Octet rule – goal is to obtain a full outer shell
    - Determines what type of bonds form
    - Nobel/inert gases – have full outer shells so they do not interact with other atoms

Free Radicals

- Free Radicals have an unpaired electron in its outermost shell
  - Sources include UV rays from sunlight, ozone, x-rays, pollution, cigarette smoke



(a) Oxygen molecule (O_2) (b) Superoxide free radical (O_2^-)

- Antioxidants *inactive* oxygen-derived free radicals

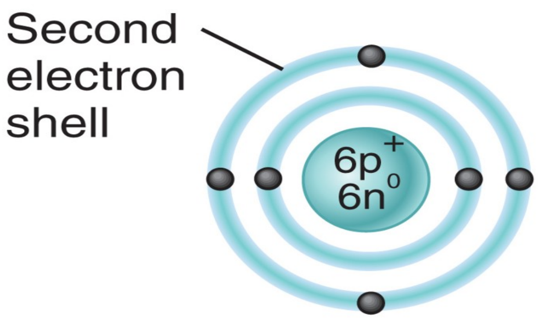
Chemical Bonds

Chemical Bonding Interactions Animation:

[Chemical Bonds]

Chemical Bonds

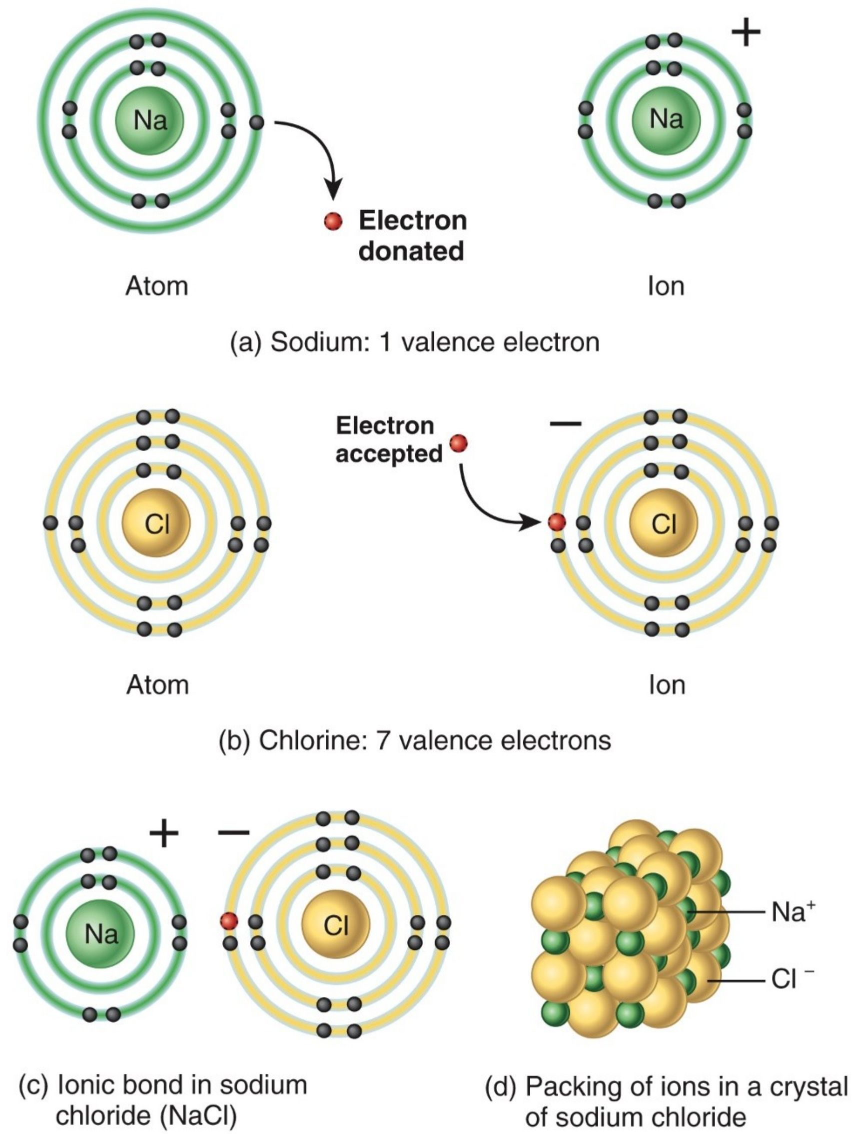
- A chemical bond occurs when atoms are held together by forces of attraction
  - The number of electrons in the valence shell determines the likelihood that an atom will form a chemical bond with another atom



Carbon (C)
Atomic number = 6
Mass number = **12** or 13
Atomic mass = 12.01

Ionic Bonds

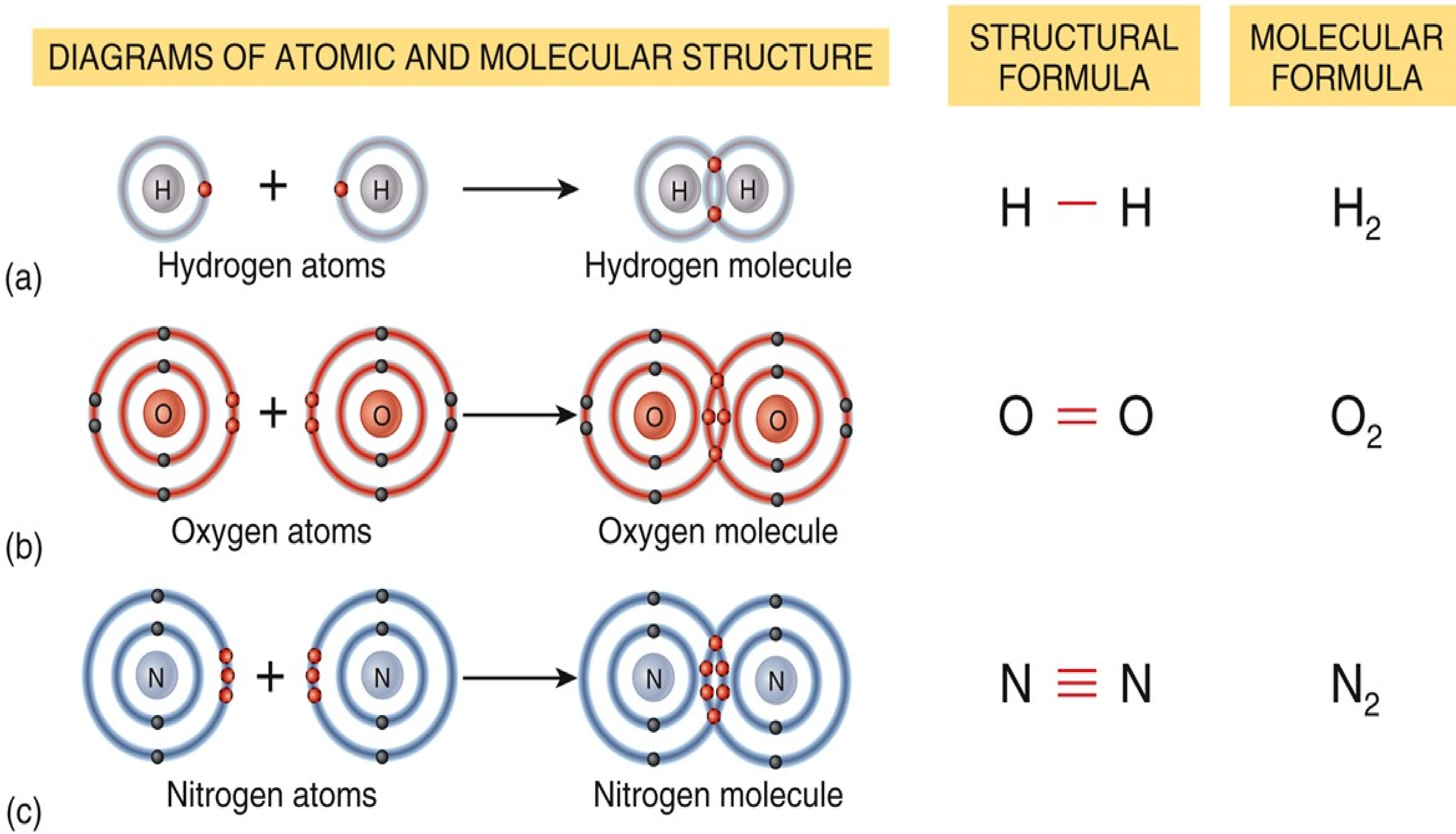
- Transfer of electrons
  - One atom donates an electron to become a positively charged cation
  - One atom accepts an electron to become a negatively charged anion
- Opposite charges attract



Covalent Bonds

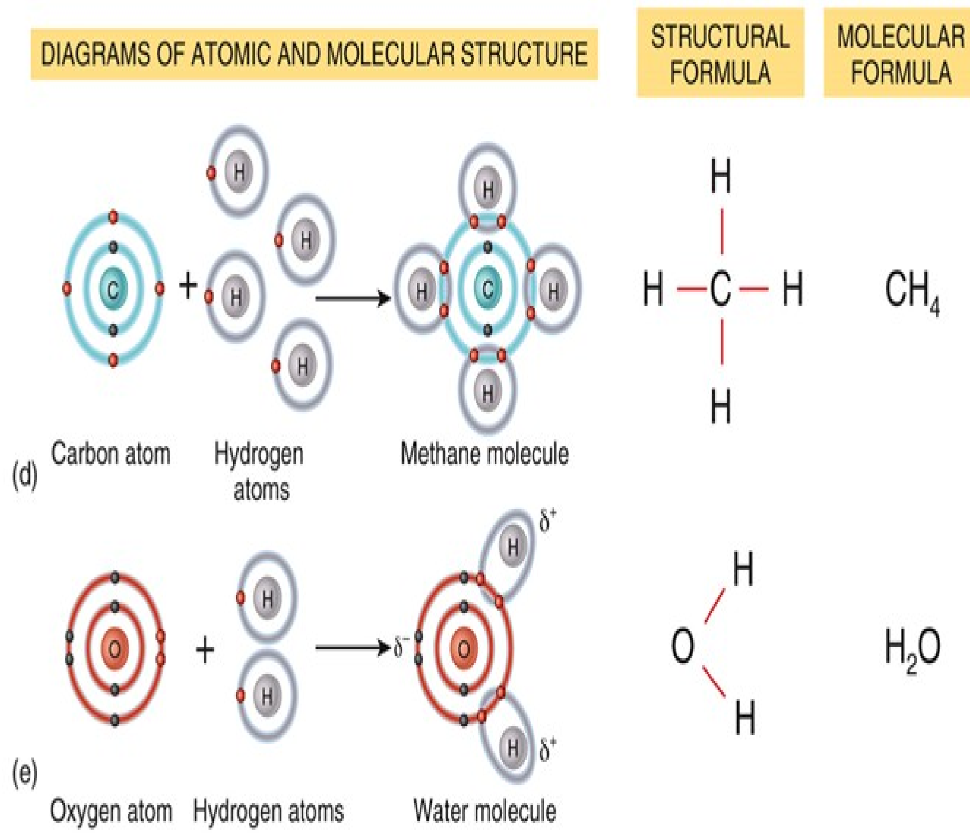
- The sharing of electron pairs
- Single bond
  - Shares one electron pair
  - Indicated by a single, horizontal line
- Double bond
  - Shares two electron pairs
  - Indicated by two horizontal lines
- Triple bond
  - Shares three electron pairs
  - Indicated by three horizontal lines

Covalent Bonds



Covalent Bonds

- Structural formula
  - Indicates specific bond arrangement
- Molecular formula
  - Indicates number and type of each atom

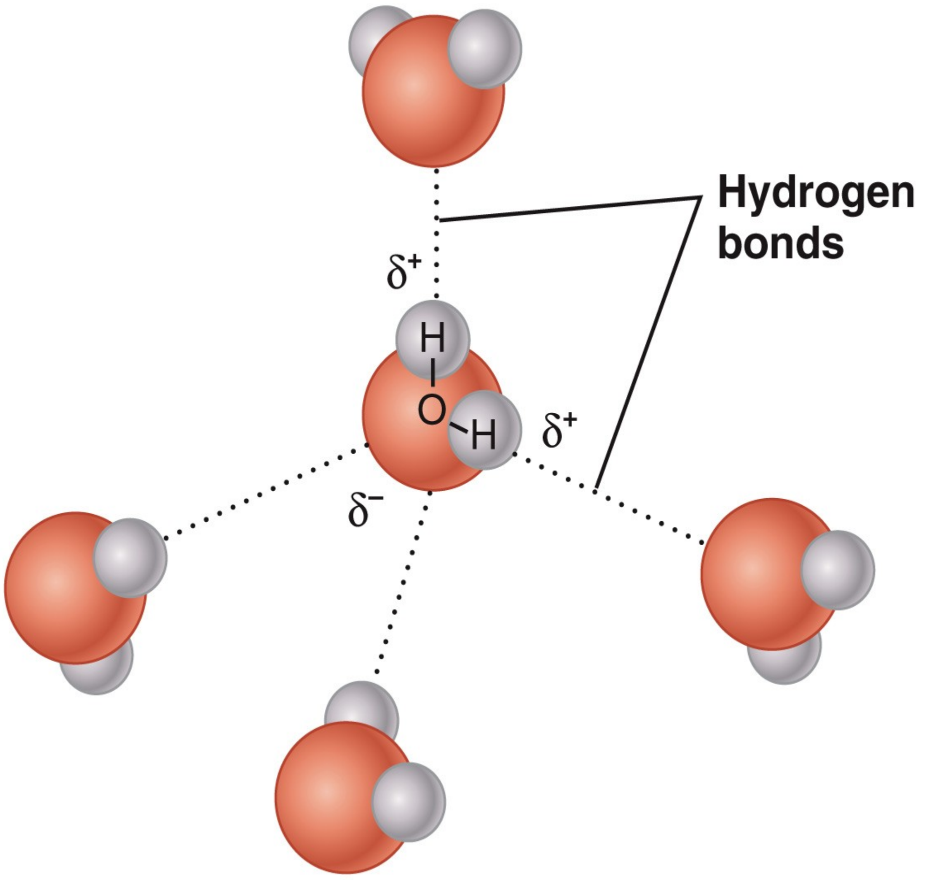


Covalent Bonds

- Polar
  - Unequal sharing of electrons
    - Molecule “side” with electrons is partially negative; “side” waiting for electrons is partially positive
- Nonpolar
  - Equal sharing of electrons
  - No partial charges associated with nonpolar molecules

Hydrogen Bonds

- Hydrogen bonds result from attraction of oppositely charged parts of molecules
- Indicated as a dotted line, since the bonds constantly break and reform



Hydrogen Bonds and Water

- Hydrogen bonds between water molecules give water cohesion
  - Cohesion is the tendency of like particles to stay together

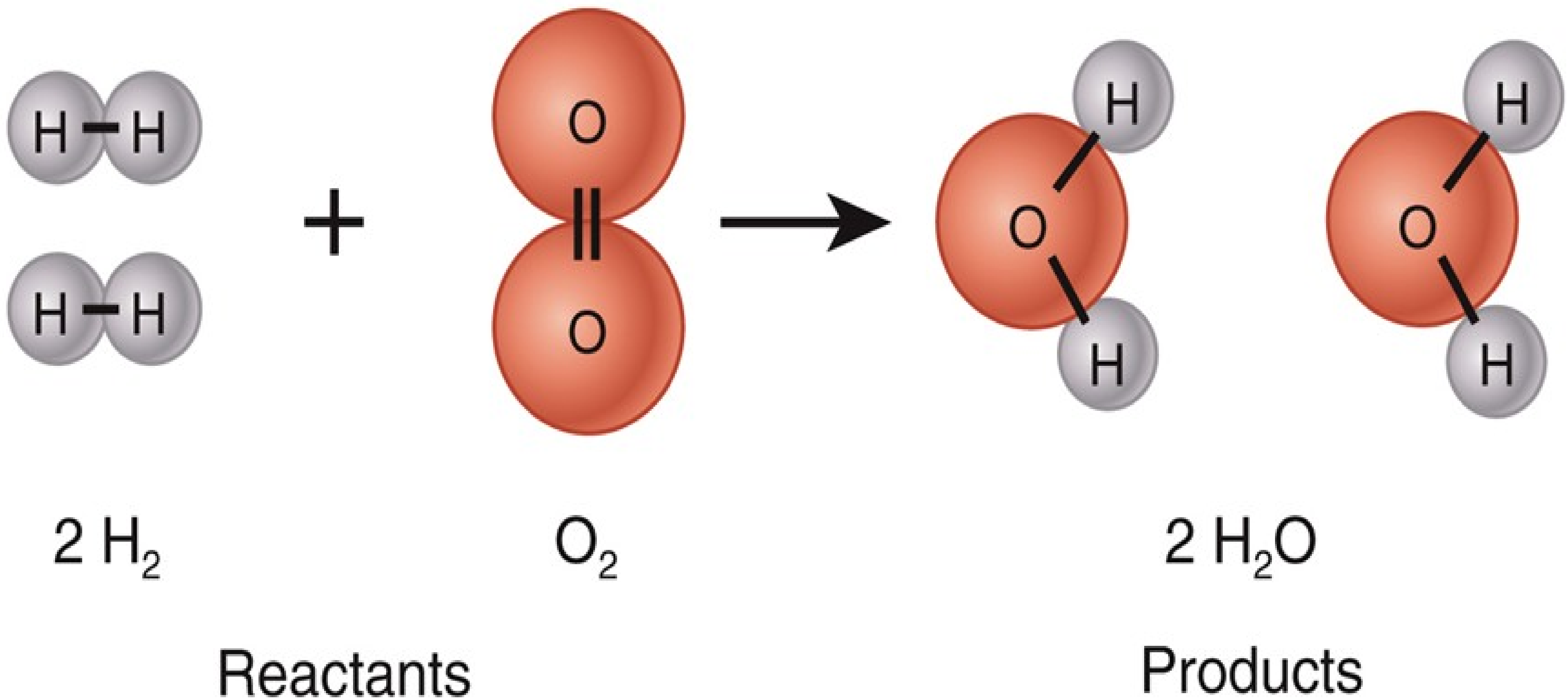
- Hydrogen bonds create surface tension
  - Surface tension is a measure of the difficulty of stretching or breaking the surface of a liquid

Chemical Reactions

Chemical Reactions

- Chemical reactions occur when new bonds are formed or old bonds are broken
  - Reactants – starting substances
  - Products – ending substances
  - Metabolism – all reactions in a body
    - Catabolic reactions
      - Releases energy by breaking down larger molecules into smaller ones
    - Anabolic reactions
      - Requires energy to form larger molecules from smaller ones

Chemical Reactions



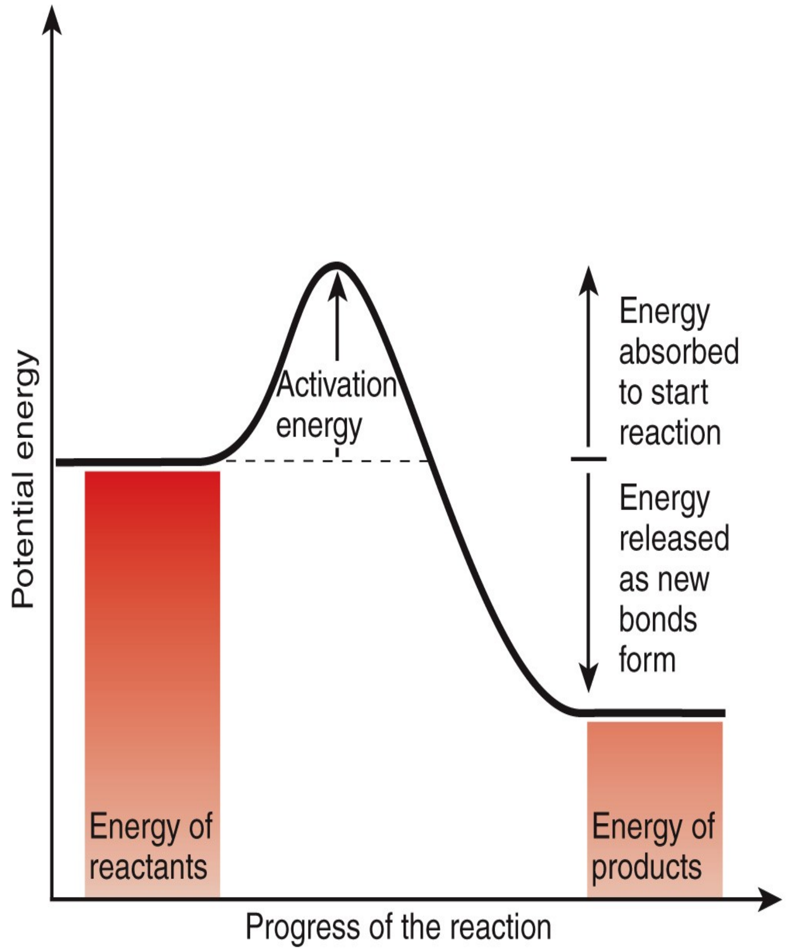
Forms of Energy & Chemical Reactions

- Energy is the capacity to do work
  - Potential energy
  - Kinetic energy
  - Chemical energy

- Law of conservation of energy – energy can neither be created nor destroyed but it can be converted from one form to another

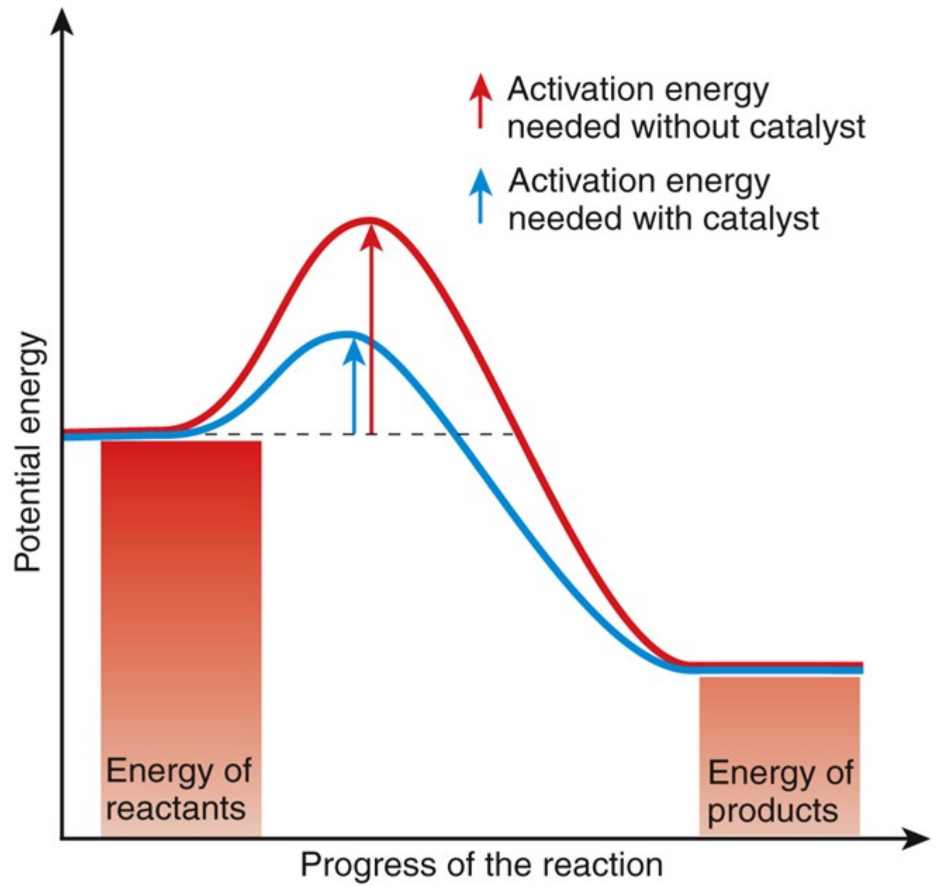
Energy Transfer

- Exergonic reactions
  - Reactants have greater energy than products
    - Releases energy
- Endergonic reactions
  - Products have greater energy than reactants
    - Requires energy
- Activation energy
  - Energy required to initially start a chemical reaction



Catalysts

- Lower activation energy requirements
- Speeds up chemical reactions
  - Enzymes are biological catalysts



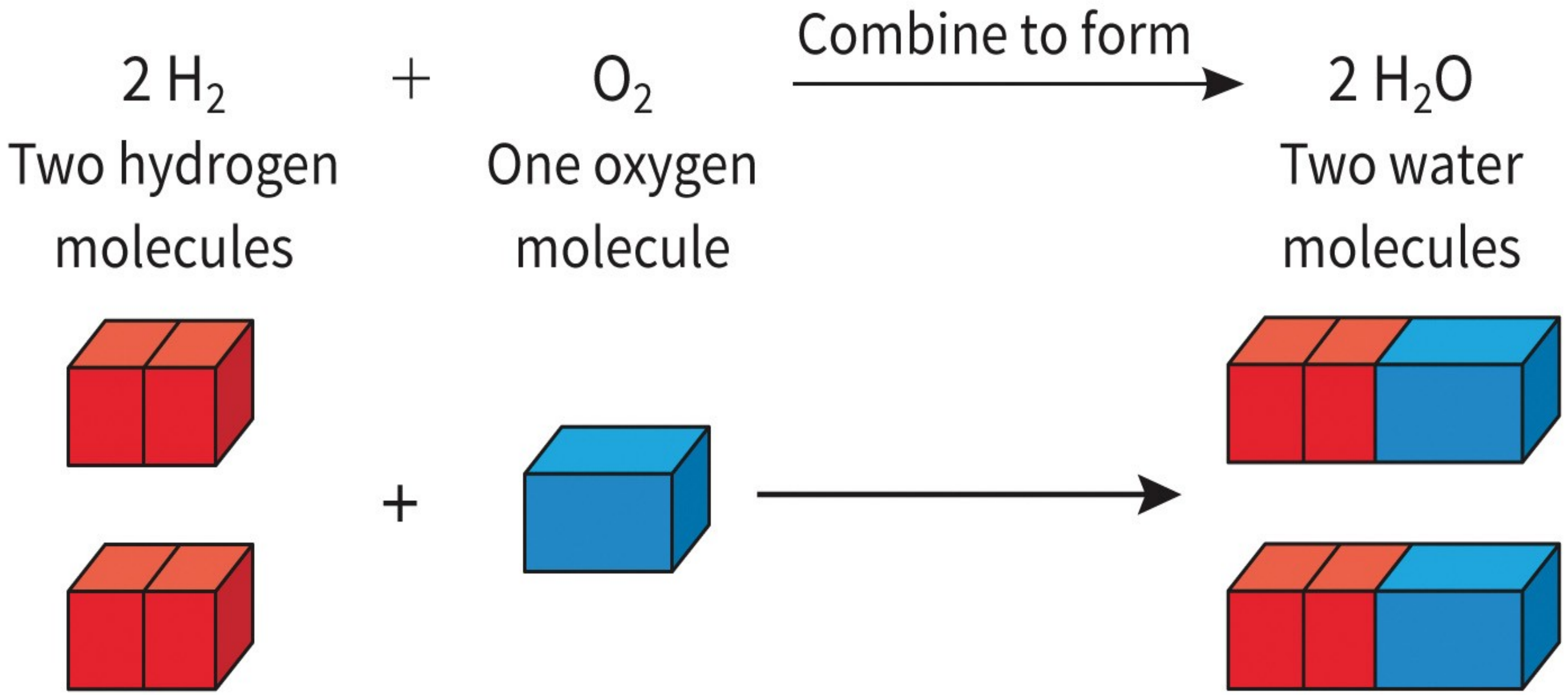
Types of Chemical Reactions

Types of Chemical Reactions

- Synthesis
- Decomposition
- Exchange
- Reversible
- Oxidation-reduction

Synthesis Reactions - Anabolism

- When two or more ions, atoms or molecules combine to form new and larger molecules

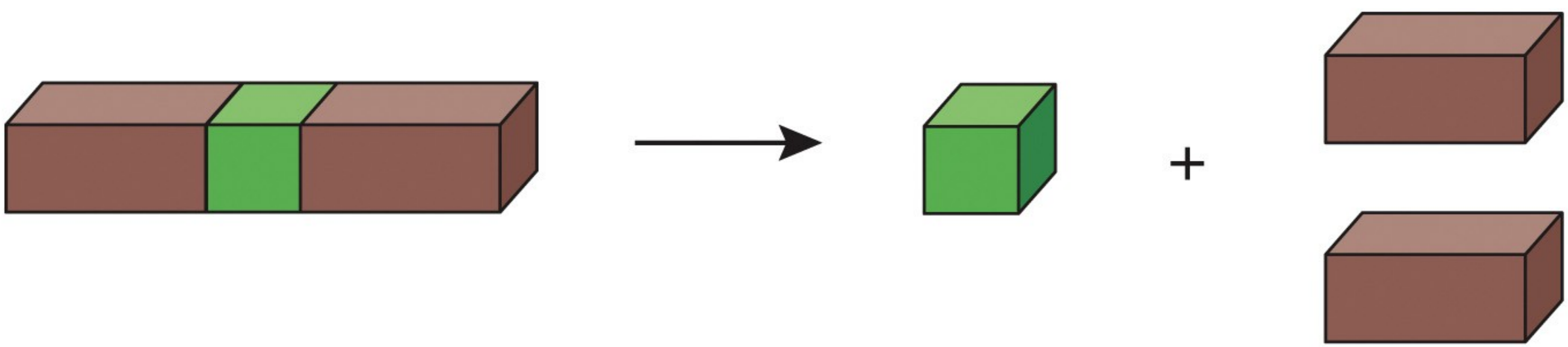


Decomposition Reactions - Catabolism

- When large molecules are split into smaller atoms, ions, atoms or molecules

$$CH_4 \xrightarrow{\text{Breaks down into}} C + 2\,H_2$$

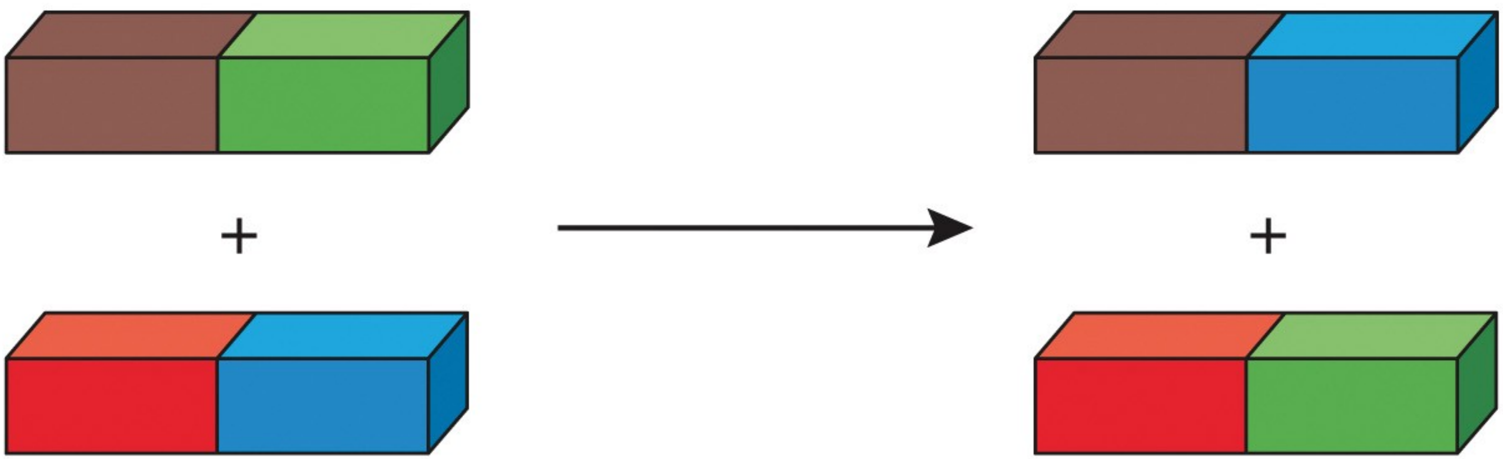
One methane molecule		One carbon atom		Two hydrogen molecules



Exchange Reactions

- Include both synthesis and decomposition reactions

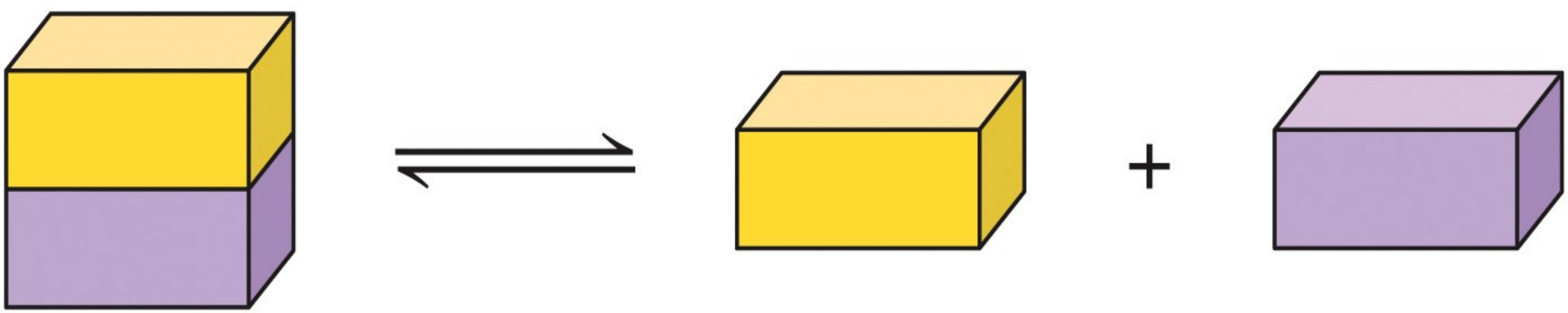
HCl	+	$NaHCO_3$	$\longrightarrow$	H_2CO_3	+	$NaCl$
Hydrochloric acid		Sodium bicarbonate		Carbonic acid		Sodium chloride



Reversible Reactions

- Can proceed in either direction, as indicated by the double arrows.
- Products can revert to original reactants

$$AB \underset{\text{Heat}}{\overset{\text{Water}}{\rightleftharpoons}} A + B$$



Oxidation-Reduction Reactions

- These reactions transfer electrons between atoms and molecules and always occur in parallel (when one substance is oxidized another is reduced)
- Oxidation – loss of electrons and energy release
- Reduction – gain of electrons and energy gain
    - Reduced form of a molecule has greater energy than its oxidized form

Inorganic Compounds and Solutions

Inorganic vs. Organic Compounds

- Inorganic compounds usually lack carbon and are simple molecules
  - Water is the most important and abundant inorganic compound in all living things

- Organic compounds always contain C and H, usually contain O, and always have covalent bonds

Properties of Water Interactions Animation:

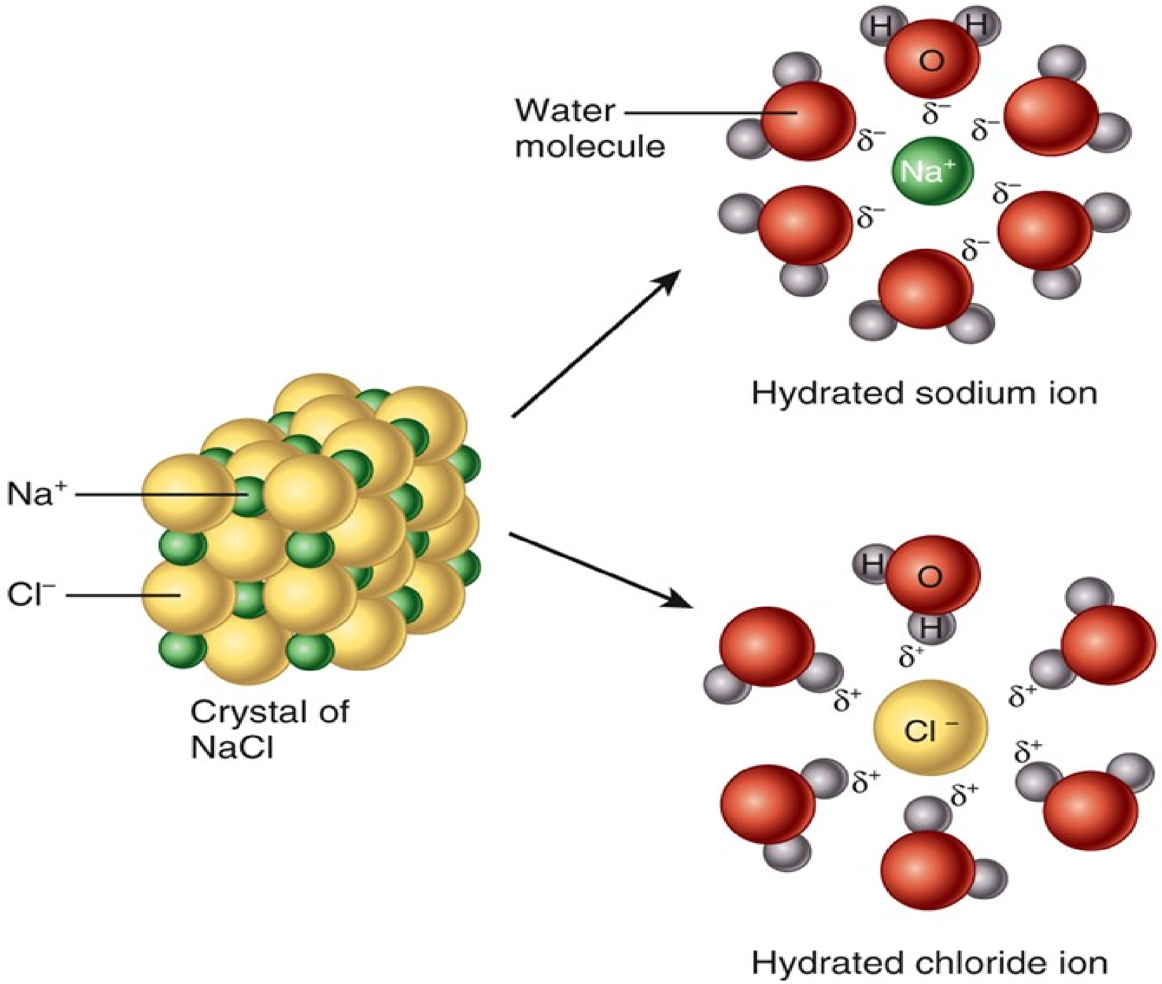
Polarity and Solubility of Molecules

Water as a Solvent

- Solution
  - Solvent dissolves the solute (another substance)
    - Biology often includes aqueous (water-based) solutions

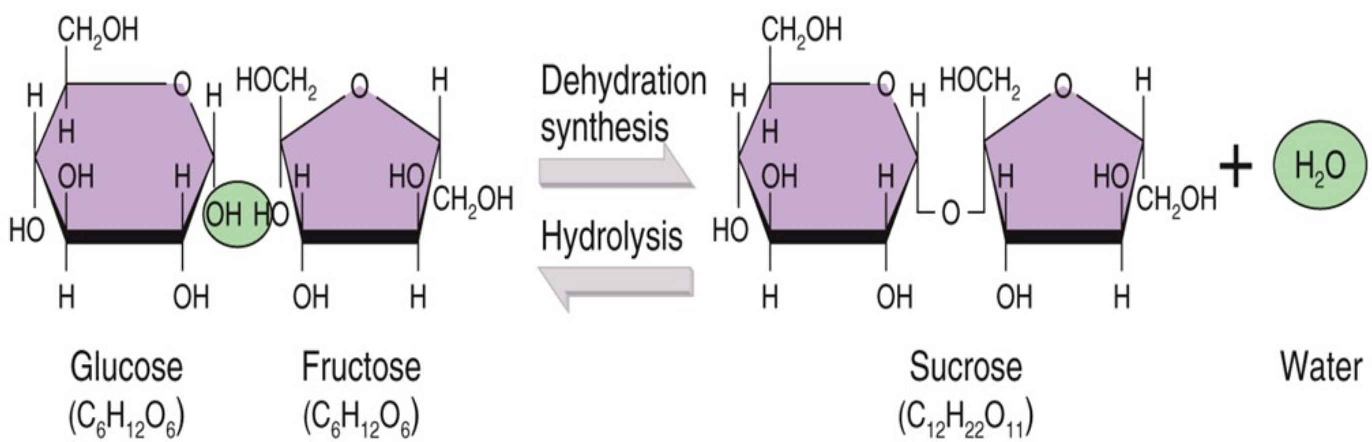
- Since water is a polar molecule, it's attracted to any other charged molecule or ion
  - Hydrophilic – "water loving"
  - Hydrophobic – "water-fearing"
    - Nonpolar molecules do not have a charge and are therefore not attracted to water

Water as a Solvent



Water in Chemical Reactions

- Water is the ideal medium
  - In a dehydration synthesis reaction, water is removed to make bonds
  - In a hydrolysis reaction, water is added to break bonds



(a) Dehydration synthesis and hydrolysis of sucrose

Thermal Properties of Water

- Water has a:
  - High heat capacity
    - Can absorb a large amount of energy without raising its temperature
      - Helps maintain human body temperature
  - High heat of vaporization
    - Water absorbs heat as it changes from a liquid to a gas
      - Evaporative cooling – when sweat, or other water, evaporates from your body, heat is removed and cools you down

Water as a Lubricant

- Water is a major component of our body fluids and helps reduce friction as membranes and organs slide over one another

Mixtures

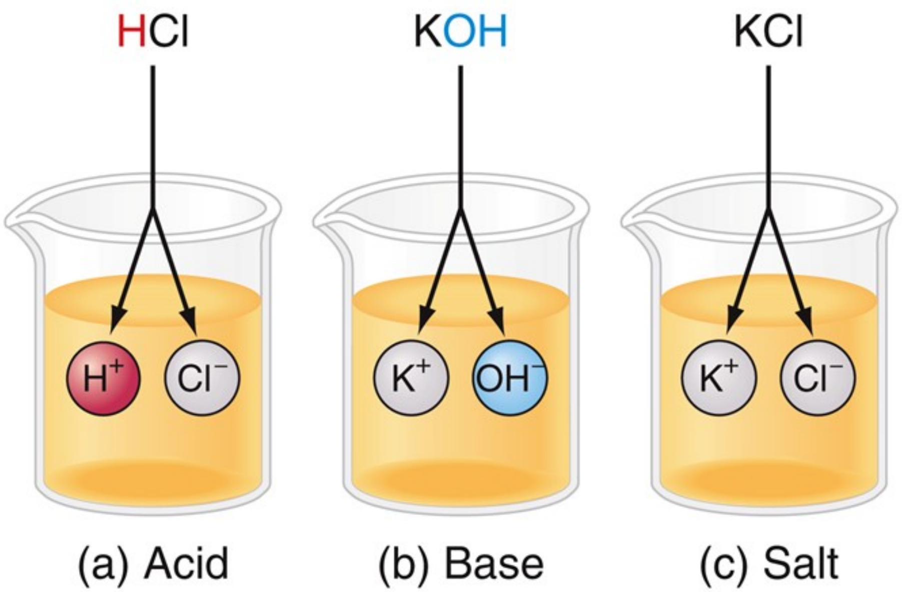
- Mixture – a combination of elements or compounds that are physically blended together but not bonded together
- 3 types of mixtures:
  - Solution
    - Once mixed together, the solutes are evenly dispersed among the solvent molecules
  - Colloid
    - Solute particles are large enough to scatter light
  - Suspension
    - Solute particles can settle out of suspending liquid

Percentage & Molarity

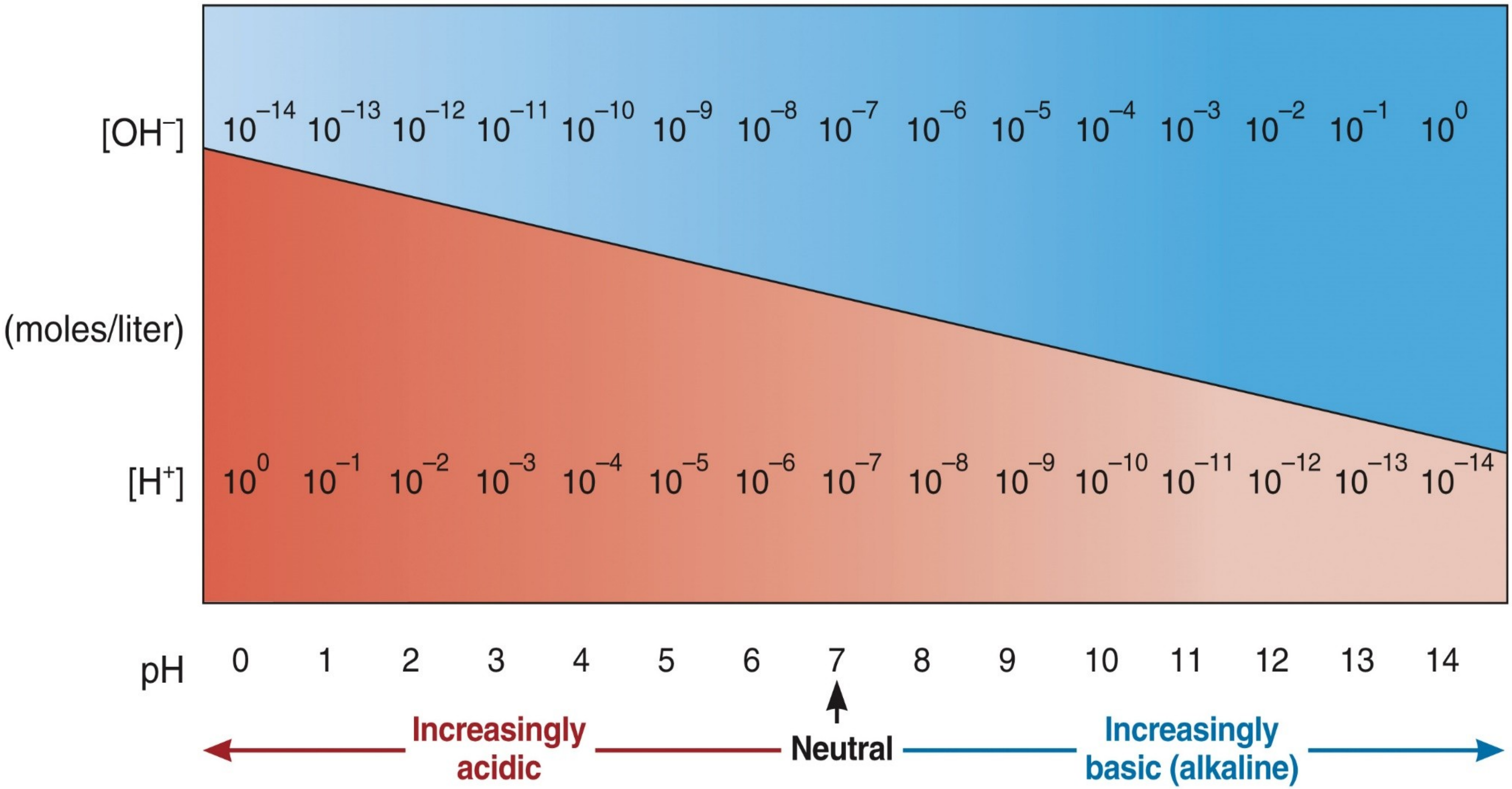
Definition	Example
Percentage (mass per volume) Number of grams of a substance per 100 milliliters (mL) of solution	To make a 10% NaCl solution, take 10 g of NaCl and add enough water to make a total of 100 mL of solution.
Molarity—(moles per liter) A 1 molar (1 M) solution = 1 mole of a solute in 1 liter of solution	To make a 1 molar (1 M) solution of NaCl, dissolve 1 mole of NaCl (58.44 g) in enough water to make a total of 1 liter of solution.

Acids, Bases, & Salts

- All dissociate in water; dissociation is separation/break down ions
  - Acids – contributes H+
  - Bases – contributes OH-
  - Salts – contributes neither H+ nor OH-



pH


[OH⁻]
10^{-14} 10^{-13} 10^{-12} 10^{-11} 10^{-10} 10^{-9} 10^{-8} 10^{-7} 10^{-6} 10^{-5} 10^{-4} 10^{-3} 10^{-2} 10^{-1} 10^{0}
(moles/liter)
[H⁺]
10^{0} 10^{-1} 10^{-2} 10^{-3} 10^{-4} 10^{-5} 10^{-6} 10^{-7} 10^{-8} 10^{-9} 10^{-10} 10^{-11} 10^{-12} 10^{-13} 10^{-14}
pH
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14
Increasingly acidic
Neutral
Increasingly basic (alkaline)

pH

- pH Values of Selected Substances (asterisks indicate substances in the human body)

Substance	pH Value
* Gastric juice (found in the stomach)	1.2–3.0
Lemon juice	2.3
Vinegar	3.0
Carbonated soft drink	3.0–3.5
Orange juice	3.5
* Vaginal fluid	3.4–4.5
Tomato juice	4.2
Coffee	5.0
* Urine	4.6–8.0
* Saliva	6.35–6.85

pH

- pH Values of Selected Substances (asterisks indicate substances in the human body)

Substance	pH Value
Milk	6.8
Distilled (pure) water	7.0
* Blood	7.35–7.45
* Semen (fluid containing sperm)	7.20–7.60
* Cerebrospinal fluid (fluid associated with nervous system)	7.4
* Pancreatic juice (digestive juice of the pancreas)	7.1–8.2
* Bile (liver secretion that aids fat digestion)	7.6–8.6
Milk of magnesia	10.5
Lye (sodium hydroxide)	14.0

pH and Buffers Interactions Animation:

[Acids and Bases]

Buffer Systems

- Maintenance of body fluid homeostasis is critical
  - $H^+ \quad + \quad HCO_3^- \leftrightarrow H_2CO_3$

    hydrogen ion   biarcarbonate ion   carbonic acid

    (weak base)
  - If blood pH drops too low, bicarbonate ion will remove excess hydrogen ions from solution, resulting in a rise of pH
  - If blood pH becomes too high, carbonic acid will dissociate to release hydrogen ions, resulting in a drop in pH

Overview of Organic Compounds

Carbon

- Organic compounds always contain carbon
- Carbons can combine in a variety of shapes
  - Carbon has 4 valence electrons, so needs to form 4 covalent bonds in order to complete its outer shell
- Carbon compounds do not dissolve easily in water
- Carbon compounds are a good source of energy

Major Functional Groups of Organic Compounds

TABLE 2.5 **Major Functional Groups of Organic Molecules**

Name and Structural Formula*	Occurrence and Significance
Hydroxyl R—O—H	*Alcohols* contain an —OH group, which is polar and hydrophilic due to its electronegative O atom. Molecules with many —OH groups dissolve easily in water.
Sulfhydryl R—S—H	*Thiols* have an —SH group, which is polar and hydrophilic due to its electronegative S atom. Certain amino acids (for example, cysteine) contain —SH groups, which help stabilize the shape of proteins.
Carbonyl $\mathrm{R}-\overset{\overset{\displaystyle \mathrm{O}}{\|}}{\mathrm{C}}-\mathrm{R}$ or $\mathrm{R}-\overset{\overset{\displaystyle \mathrm{O}}{\|}}{\mathrm{C}}-\mathrm{H}$	*Ketones* contain a carbonyl group within the carbon skeleton. The carbonyl group is polar and hydrophilic due to its electronegative O atom. *Aldehydes* have a carbonyl group at the end of the carbon skeleton.

*R = variable group.

Major Functional Groups of Organic Compounds

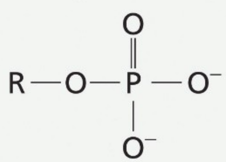
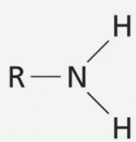
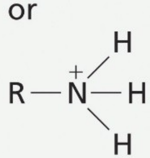
TABLE 2.5 Major Functional Groups of Organic Molecules

Name and Structural Formula*	Occurrence and Significance
Carboxyl $R-\overset{\overset{\displaystyle O}{\Vert}}{C}-OH$ or $R-\overset{\overset{\displaystyle O}{\Vert}}{C}-O^{-}$	*Carboxylic acids* contain a carboxyl group at the end of the carbon skeleton. All amino acids have a —COOH group at one end. The negatively charged form predominates at the pH of body cells and is hydrophilic.
Ester $R-\overset{\overset{\displaystyle O}{\Vert}}{C}-O-R$	*Esters* predominate in dietary fats and oils and also occur in our body as triglycerides. Aspirin is an ester of salicylic acid, a pain-relieving molecule found in the bark of the willow tree.

*R = variable group.

Major Functional Groups of Organic Compounds

TABLE 2.5 Major Functional Groups of Organic Molecules

Name and Structural Formula*	Occurrence and Significance
Phosphate 	*Phosphates* contain a phosphate group (—PO_4^{2-}), which is very hydrophilic due to the dual negative charges. An important example is adenosine triphosphate (ATP), which transfers chemical energy between organic molecules during chemical reactions.
Amino  or 	*Amines* have an —NH_2 group, which can act as a base and pick up a hydrogen ion, giving the amino group a positive charge. At the pH of body fluids, most amino groups have a charge of 1^+. All amino acids have an amino group at one end.

*R = variable group.

Macromolecules

- Carbohydrates
- Lipids
- Proteins
- Nucleic acids

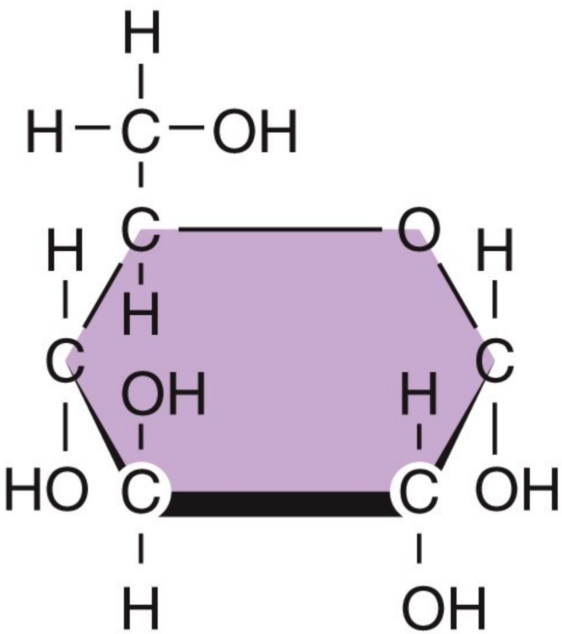
Carbohydrates

- Comprised of carbon, hydrogen and oxygen
- Include sugars, glycogen, starches and cellulose
- Main source of chemical energy
- 2-3% of total body mass

Major Carbohydrate Groups

Type of Carbohydrate	Examples
Monosaccharides (simple sugars that contain from 3 to 7 carbon atoms)	Glucose (the main blood sugar). Fructose (found in fruits). Galactose (in milk sugar). Deoxyribose (in DNA). Ribose (in RNA).
Disaccharides (simple sugars formed from the combination of two monosaccharides by dehydration synthesis)	Sucrose (table sugar) = glucose + fructose. Lactose (milk sugar) = glucose + galactose. Maltose = glucose + glucose.

Glucose



All atoms written out

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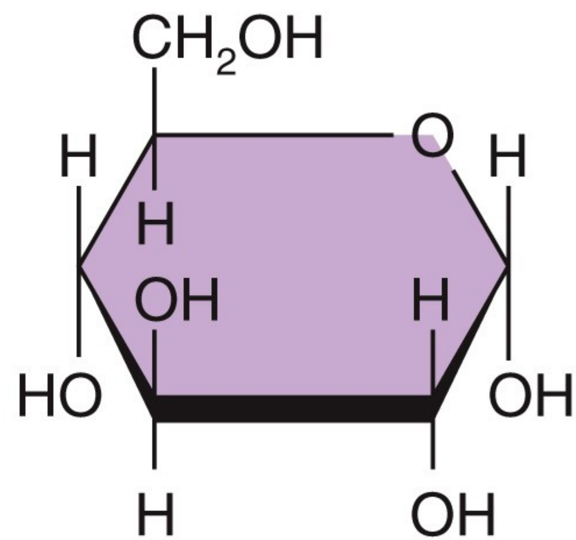
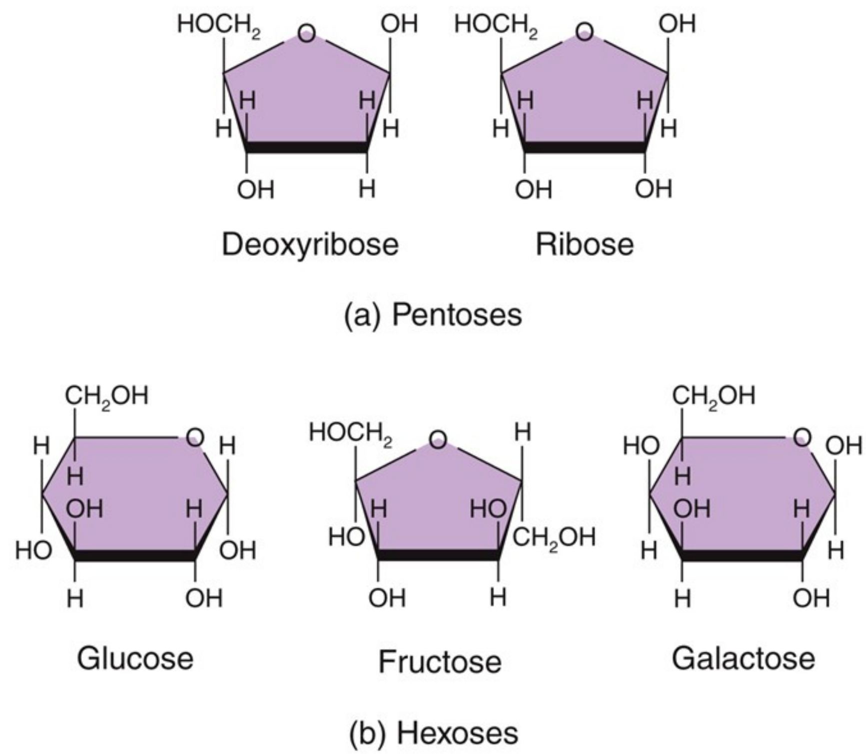


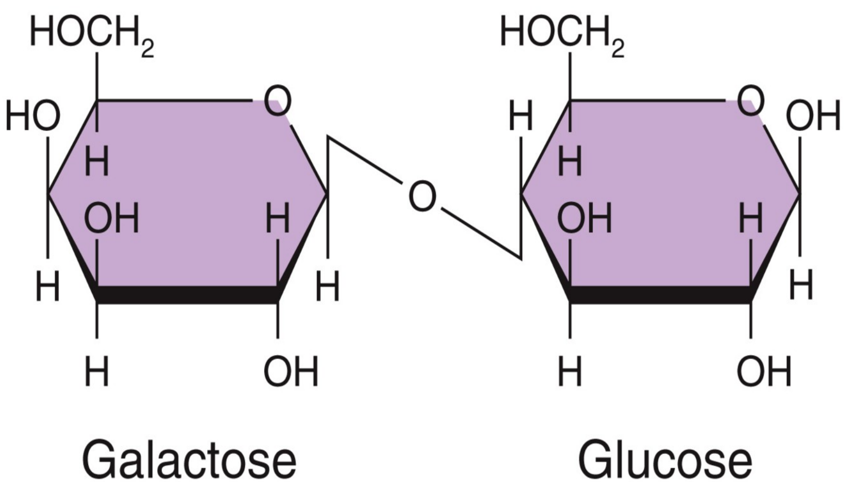
Standard shorthand

Monosaccharides

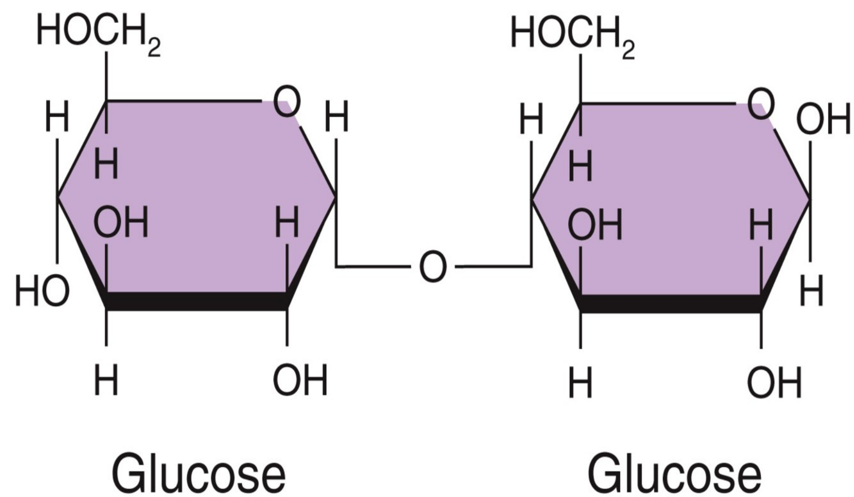
- Ratio of 1:2:1 of carbon:hydrogen:oxygen
  - Isomers – have same molecular formula but vary in structural formula
  - Pentoses
    - 5-carbon sugars
    - Often used as a component of DNA and RNA
  - Hexoses
    - 6-carbon sugars
    - Often used for energy



Disaccharides



(b) Lactose



(c) Maltose

- 2 monosaccharides joined by dehydration synthesis
- Both monosaccharide and disaccharide names tend to end with "-ose"

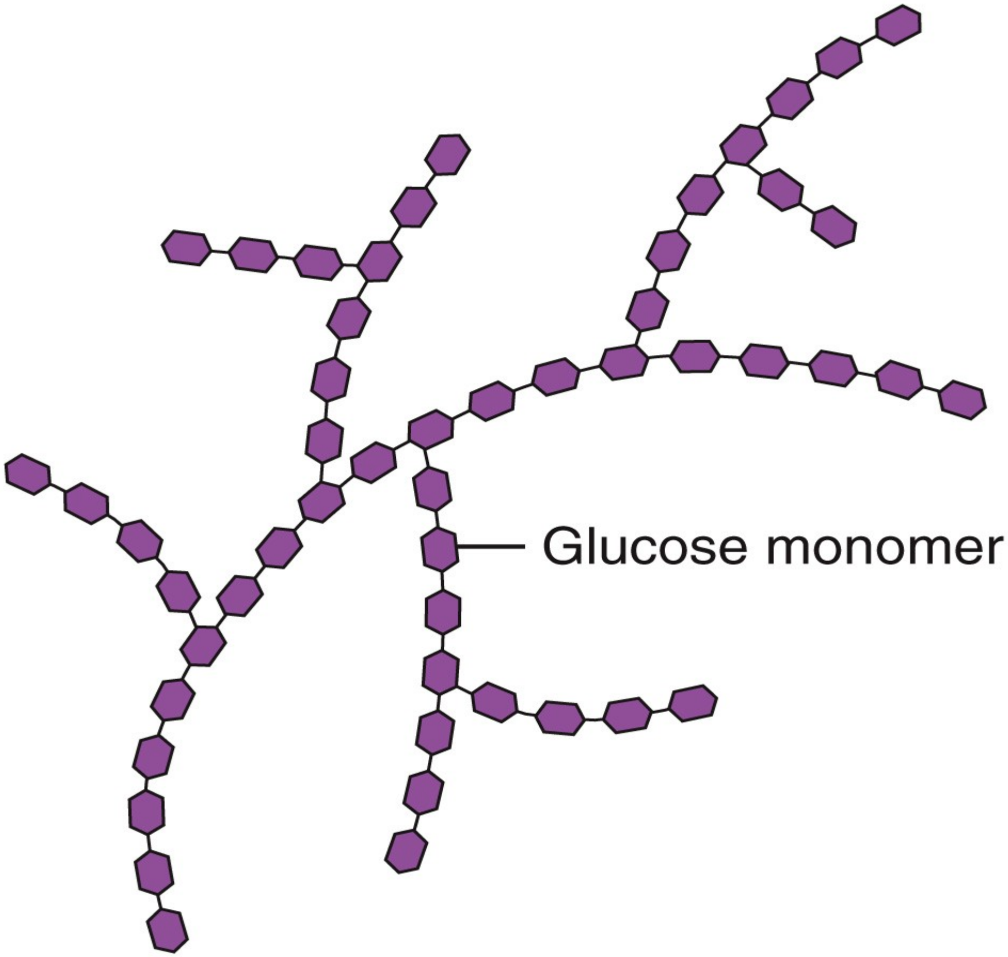
Major Carbohydrate Groups

Type of Carbohydrate	Examples
Polysaccharides (from tens to hundreds of monosaccharides joined by dehydration synthesis)	Glycogen (stored form of carbohydrates in animals). Starch (stored form of carbohydrates in plants and main carbohydrates in food). Cellulose (part of cell walls in plants that cannot be digested by humans but aids movement of food through intestines).

Carbohydrates

- Polysaccharides/complex carbohydrates
- Storage carbohydrates
  - Glycogen – stored in liver and muscle
    - Will hydrolyze into its glucose subunits in response to drop in blood sugar
  - Starch – plant equivalent to glycogen
- Structural carbohydrates
  - Cellulose – part of plant cell walls; cannot be digested by humans but aids with movement of GI tract contents

Polysaccharides



Types of Lipids in the Body

Type of Lipid	Functions
Fatty acids	Used to synthesize triglycerides and phospholipids or catabolized to generate adenosine triphosphate (ATP).
Triglycerides (fats and oils)	Protection, insulation, energy storage.
Phospholipids	Major lipid component of cell membranes.

Types of Lipids in the Body

Type of Lipid	Functions
Steroids: Cholesterol	Minor component of all animal cell membranes; precursor of bile salts, vitamin D, and steroid hormones.
Steroids: Bile salts	Needed for digestion and absorption of dietary lipids.
Steroids: Vitamin D	Helps regulate calcium level in body; needed for bone growth and repair.
Steroids: Adrenocortical hormones	Help regulate metabolism, resistance to stress, and salt and water balance.
Steroids: Sex hormones	Stimulate reproductive functions and sexual characteristics.

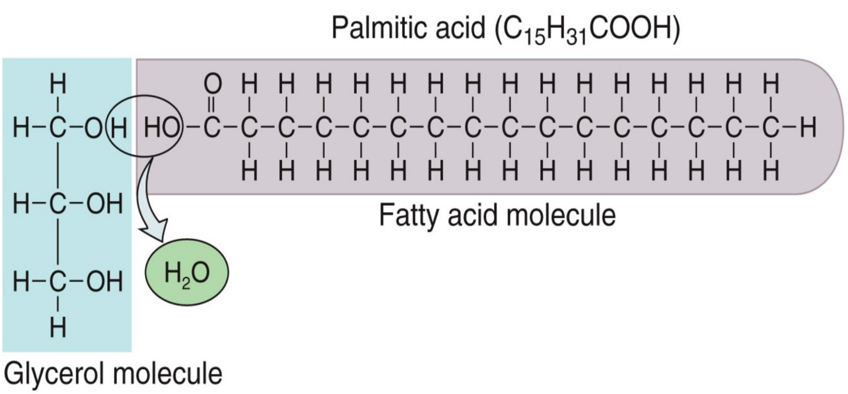
Types of Lipids in the Body

Type of Lipid	Functions
Eicosanoids (prostaglandins and leukotrienes)	Have diverse effects on modifying responses to hormones, blood clotting, inflammation, immunity, stomach acid secretion, airway diameter, lipid breakdown, and smooth muscle contraction.
Other lipids: Carotenes	Needed for synthesis of vitamin A (used to make visual pigments in eye); function as antioxidants.
Other lipids: Vitamin E	Promotes wound healing, prevents tissue scarring, contributes to normal structure and function of nervous system, and functions as antioxidant.
Other lipids: Vitamin K	Required for synthesis of blood-clotting proteins.
Other lipids: Lipoproteins	Transport lipids in blood, carry triglycerides and cholesterol to tissues, and remove excess cholesterol from blood.

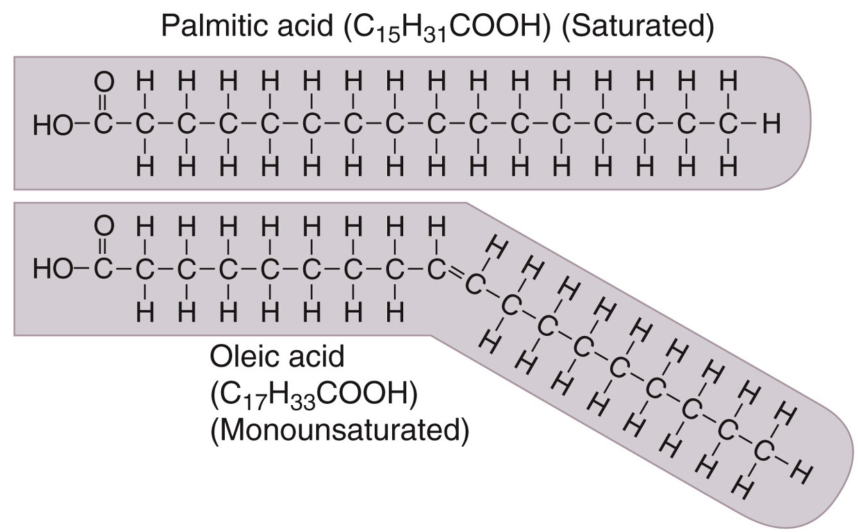
Fatty Acids

- Fatty acids can be saturated or unsaturated
- Saturated fatty acids (no double bonds)
  - Linear
  - Solid at room temperature
- Unsaturated fatty acids
  - Bends at double-bond location
  - Liquid at room temperature

Fatty Acids



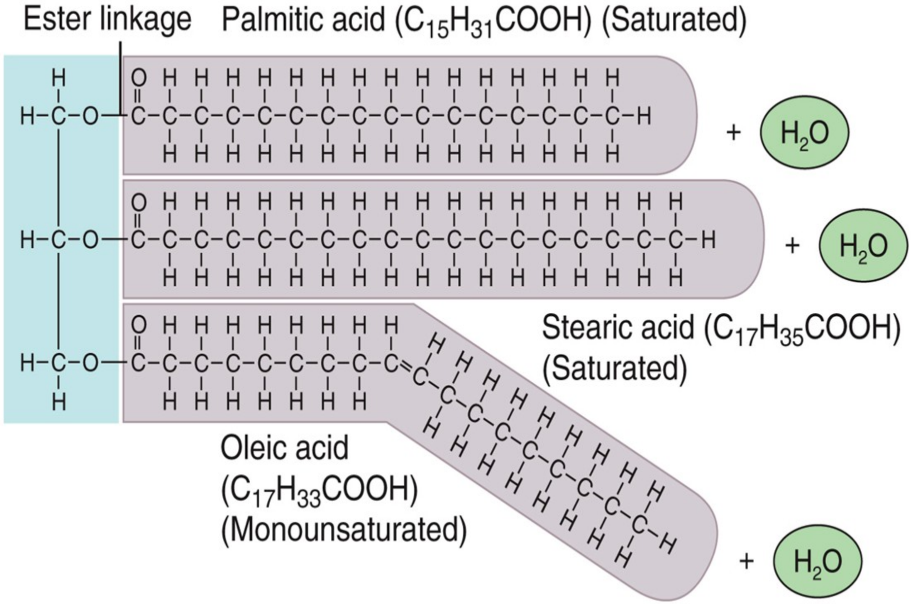
(b) Dehydration synthesis involving glycerol and a fatty acid



(a) Structures of saturated and unsaturated fatty acids

Triglycerides

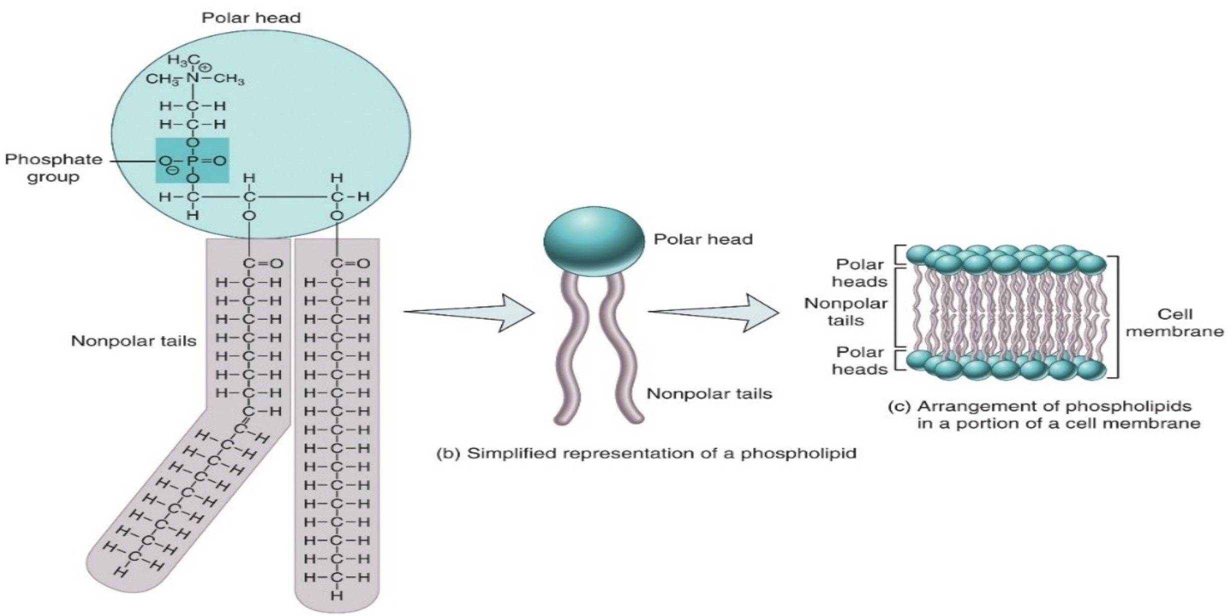
- Also known as fats
- Glycerol with 3 fatty acid chains
- Triglycerides provide protection, insulation, and energy



(c) Triglyceride (fat) molecule

Phospholipids

- Phospholipids are an important component of cell membranes
- Glycerol, phosphate, with two fatty acid tails
- Hydrophilic heads; hydrophobic tails
    - Forms micelles (droplets) or bilayers in effort to isolate tails from water-based/aqueous solutions

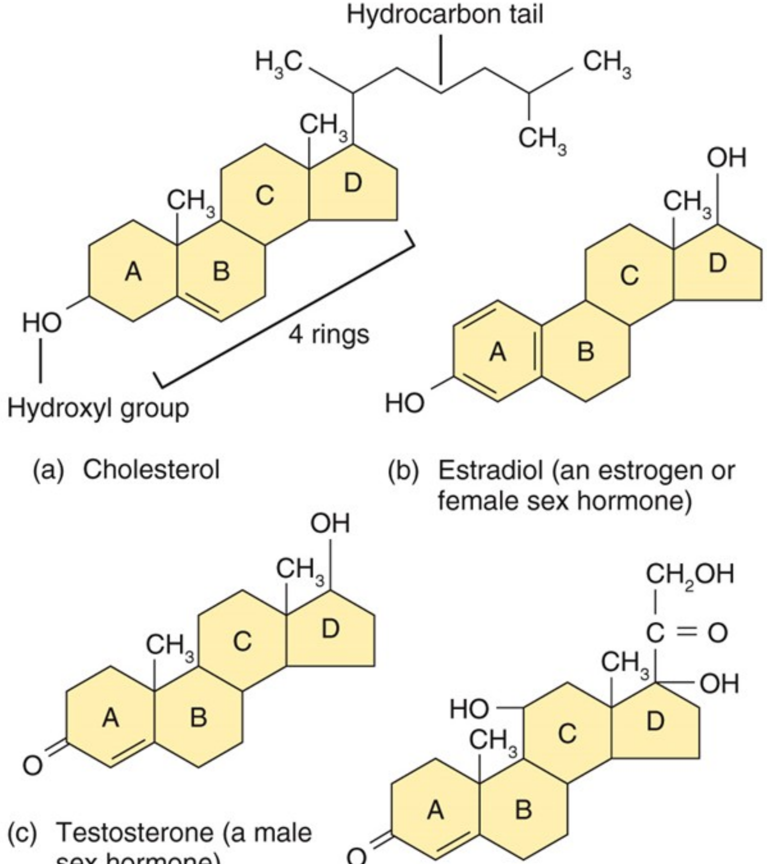


(a) Chemical structure of a phospholipid
(b) Simplified representation of a phospholipid
(c) Arrangement of phospholipids in a portion of a cell membrane

Steroids

- 4 fused rings with various groups attached

- Cholesterol is the precursor to sex hormones
  - Cholesterol is also component of plasma membrane



(a) Cholesterol

(b) Estradiol (an estrogen or female sex hormone)

(c) Testosterone (a male sex hormone)

(d) Cortisol

Proteins

- Proteins give structure to the body, regulate processes, provide protection, assist in muscle contraction, transport substances, and serve as enzymes

Functions of Proteins

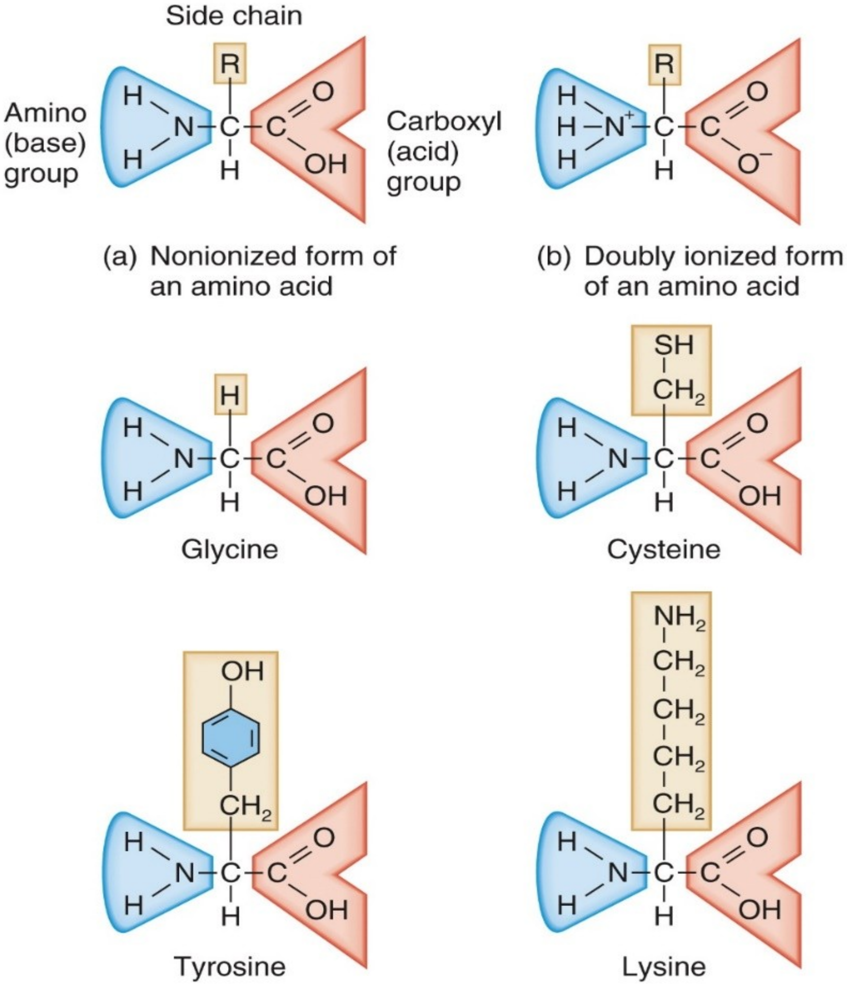
Type of Protein	Functions
Structural	Form structural framework of various parts of body. *Examples*: collagen in bone and other connective tissues; keratin in skin, hair, and fingernails.
Regulatory	Function as hormones that regulate various physiological processes; control growth and development; as neurotransmitters, mediate responses of nervous system. *Examples*: the hormone insulin (regulates blood glucose level); the neurotransmitter known as substance P (mediates sensation of pain in nervous system).
Contractile	Allow shortening of muscle fibers (cells) which produces movement. *Examples*: myosin; actin.

Functions of Proteins

Type of Protein	Functions
Immunological	Aid responses that protect body against foreign substances and invading pathogens. *Examples*: antibodies; interleukins.
Transport	Carry vital substances throughout body. *Example*: hemoglobin (transports most oxygen and some carbon dioxide in blood).
Catalytic	Act as enzymes that regulate biochemical reactions. *Examples*: salivary amylase; sucrase; ATPase.

Amino Acids

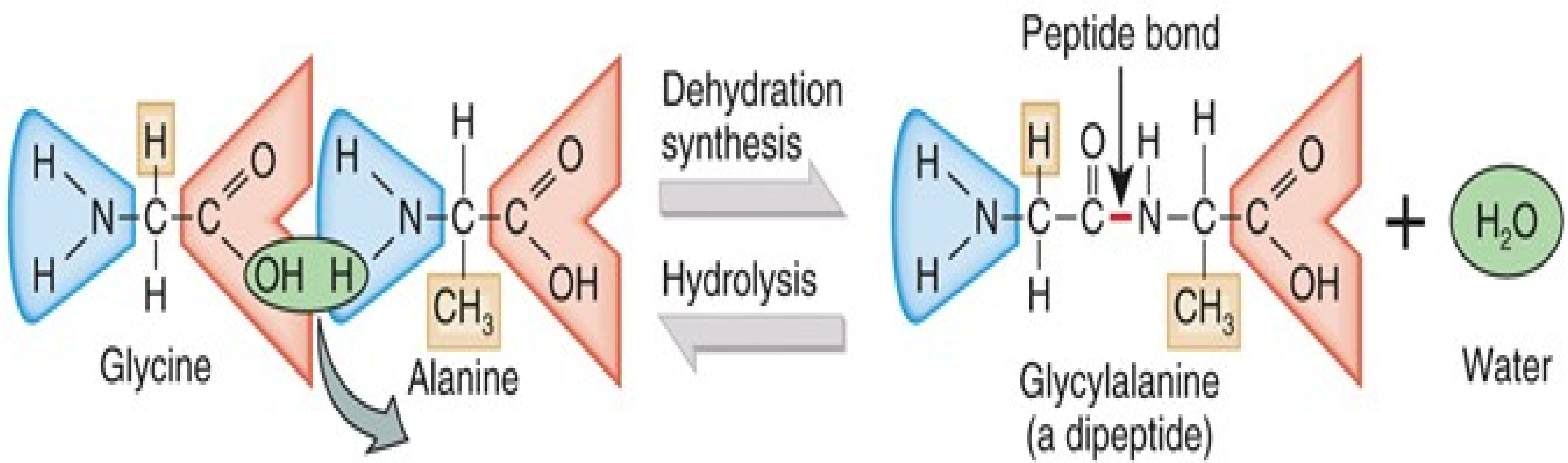
- Monomer of proteins
- Structure
  - Central carbon atom forms 4 bonds with:
    - Amino group
    - Carboxyl group
    - H atom
    - R group
      - What is at the variable R region defines the specific amino acid and its chemical properties



(a) Nonionized form of an amino acid
(b) Doubly ionized form of an amino acid
(c) Representative amino acids

Peptide Bond Formation

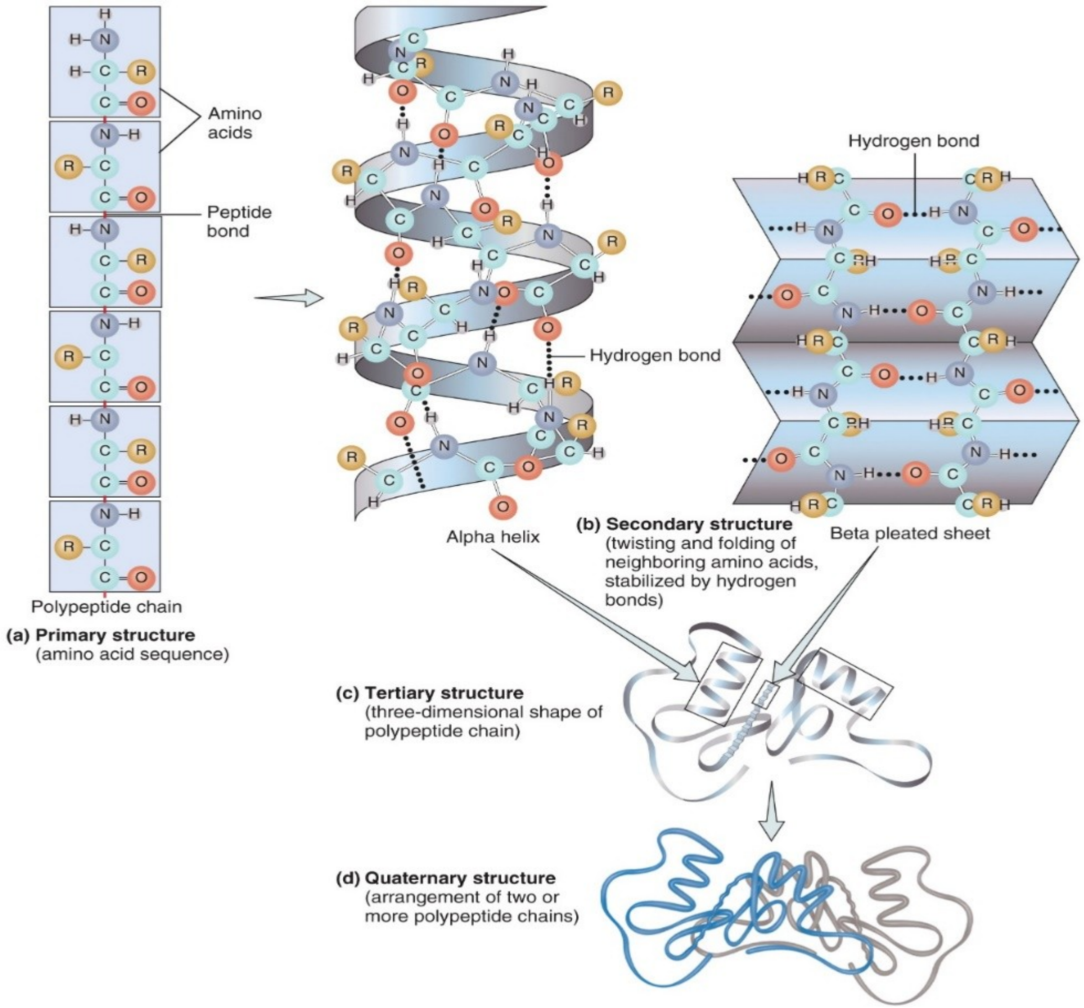
- Amino acids are linked together via dehydration synthesis
  - Peptide bond
  - Polypeptide – chain of dozens to hundreds of amino acids linked together



Protein Structure

- Primary structure
  - Sequence of amino acids in the chain
- Secondary structure
  - Formed by hydrogen bonds
    - Alpha helix or beta pleated sheath
- Tertiary structure
  - Interaction of R groups
- Quaternary structure
  - If a protein is composed of more than one polypeptide chain, how the chains join together

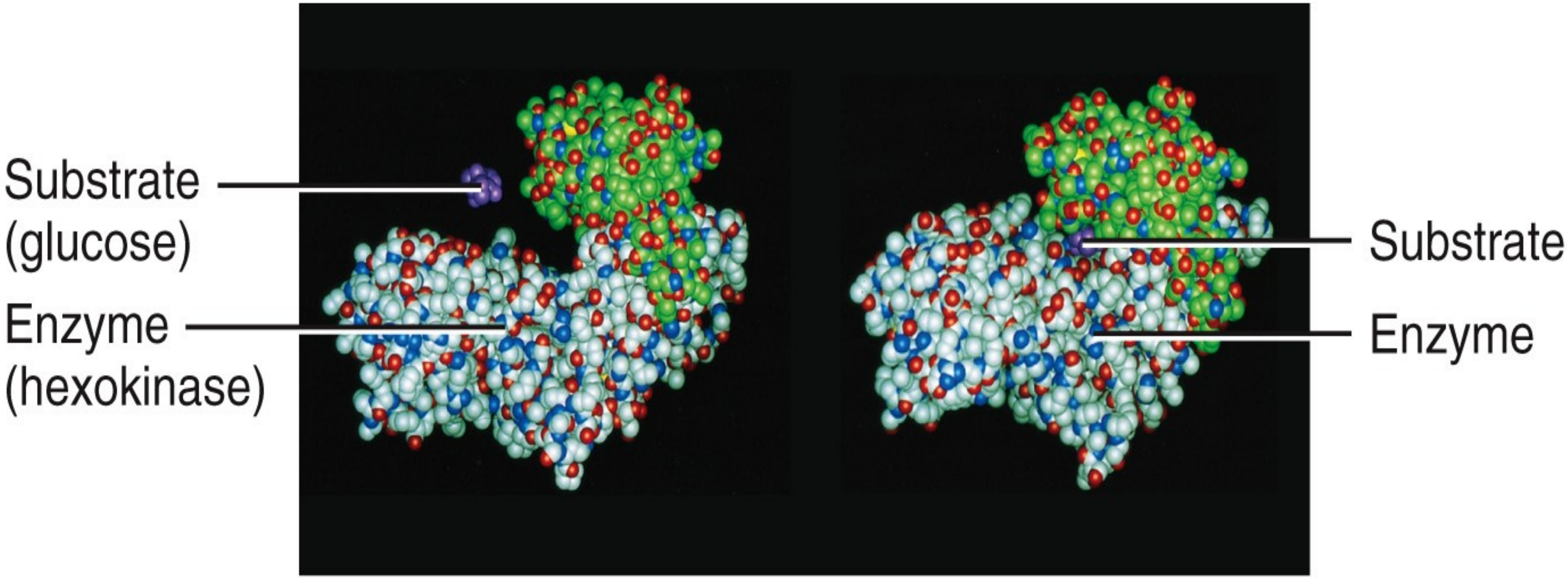
Structural Organization in Proteins



Proteins - Enzymes

- Active site – pocket/cleft where the substrate (reactant) fits
- Induced fit
  - Once the substrate binds to active site, the enzyme shape alters slightly
- An enzyme is a catalyst in a living cell
- Enzymes are:
  - Highly specific
  - Extremely efficient
  - Subject to cellular controls

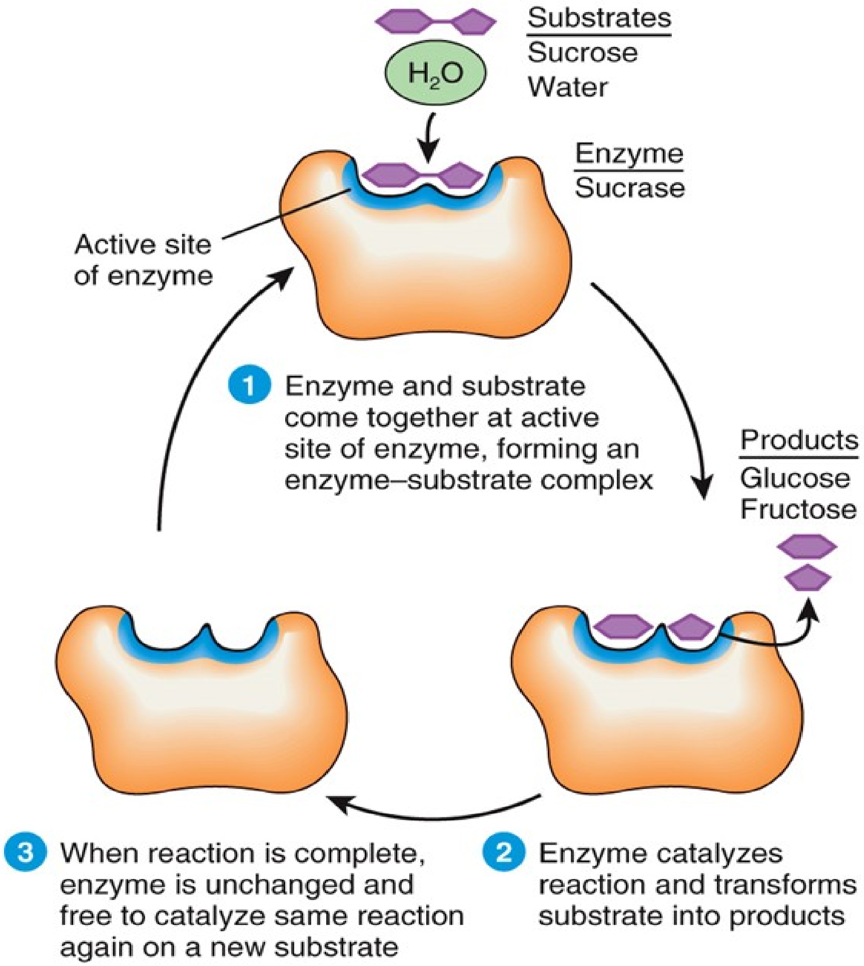
Enzyme-Substrate Complexes



(b) Molecular model of enzyme and substrate uncombined (left) and enzyme–substrate complex (right)

Enzymes

- Names tend to end in "-ase"
    - Often named by their substrate (example: sucrase {sucrose is substrate}) or their action (example: dehydrogenase – removes hydrogen)
- Are not altered; can be reused



(a) Mechanism of enzyme action

Enzymes
Interactions Animation:

[Enzyme Functions and ATP]

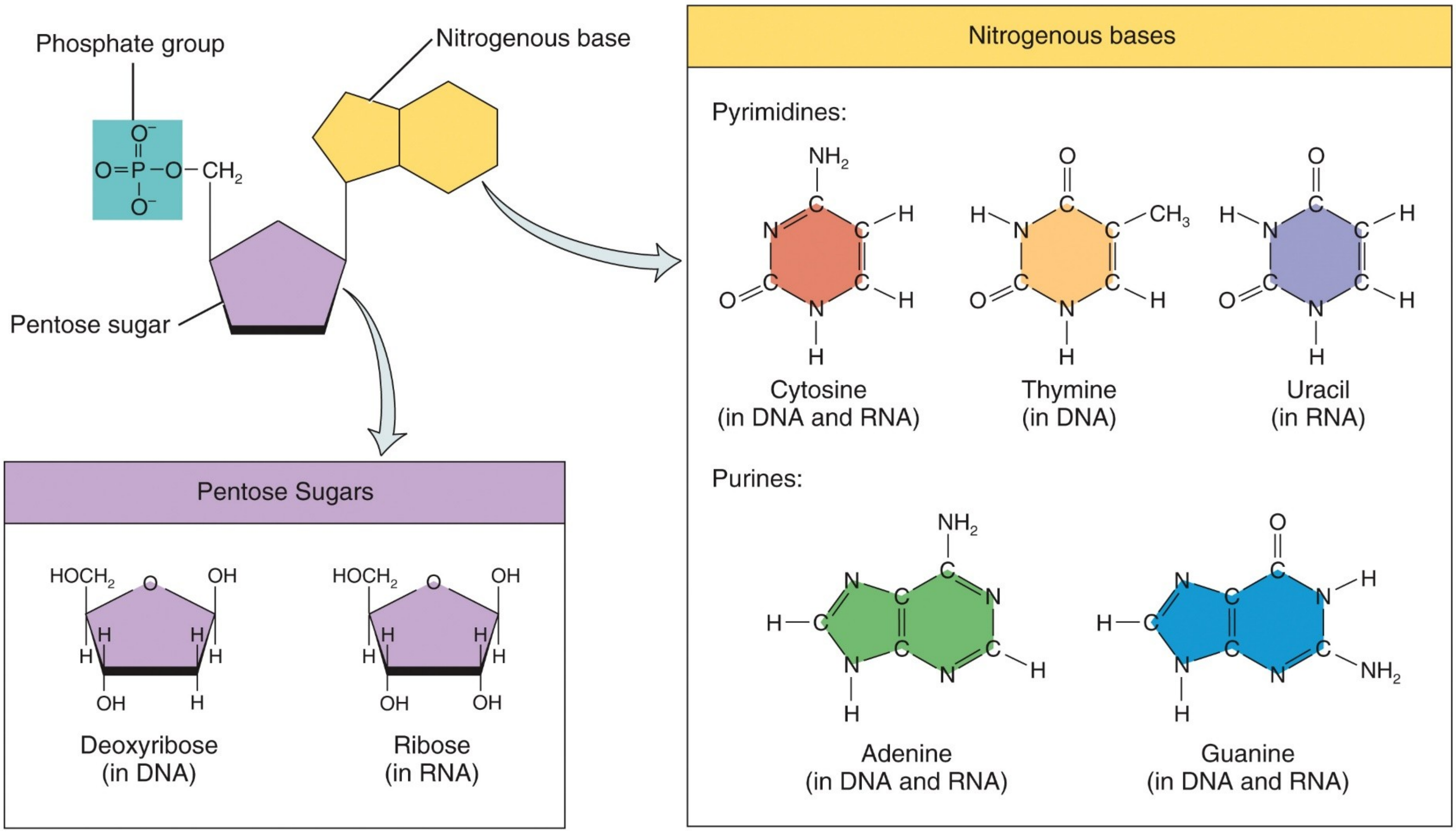
Nucleic Acids

- DNA – deoxyribonucleic acid
  - Forms the genetic code and regulates most of the cell's activities
- RNA – ribonucleic acid
  - Guides protein formation

Nucleotide

- Monomer of nucleic acids
- Structure
  - Pentose sugar
    - Deoxyribose in DNA; ribose in RNA
  - Phosphate group
  - Nitrogenous base
    - Adenine
    - Guanine
    - Cytosine
    - Thymine (only in DNA)
    - Uracil (only in RNA)

Components of a Nucleotide



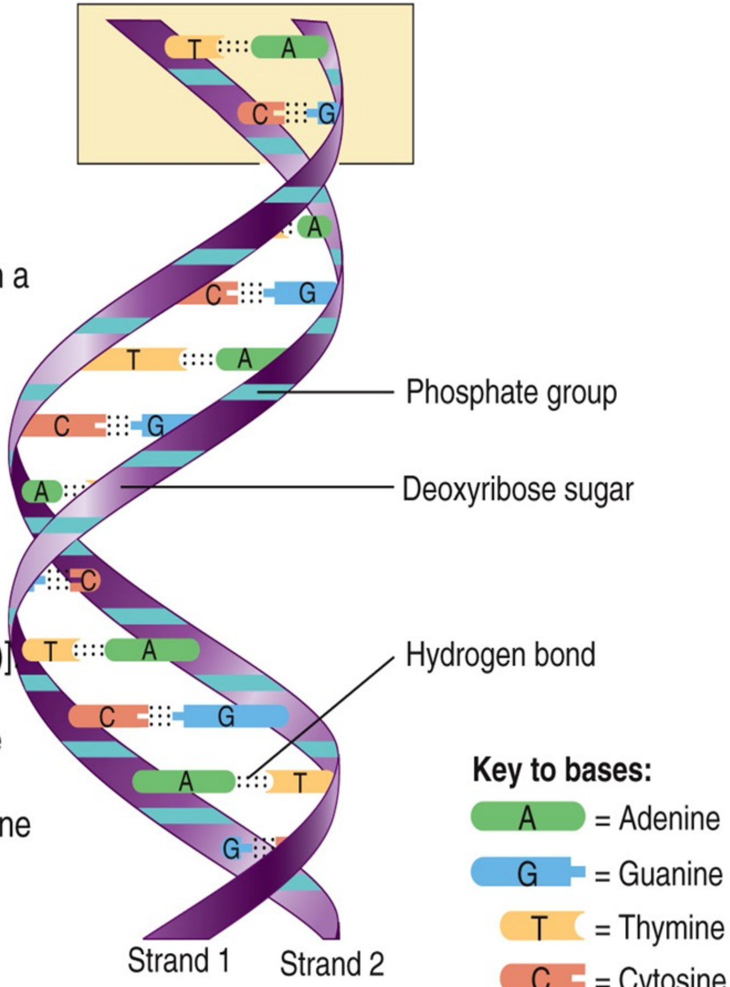
Nucleic Acids

- DNA is double-stranded
  - Adenine hydrogen bonds with thymine
  - Guanine hydrogen bonds with cytosine

- RNA is single-stranded

DNA

- DNA is made of two strands twisted in a spiral staircase-like structure called a double helix.
- Each strand consists of nucleotides bound together.
- Each nucleotide consists of a deoxyribose sugar bound to a phosphate group and one of 4 nitrogenous bases [adenine (A), thymine (T), guanine (G), cytosine (C)]
- The nitrogenous bases pair together through hydrogen bonding to form the "steps" of the double helix.
- Adenine pairs with thymine and guanine pairs with cytosine.



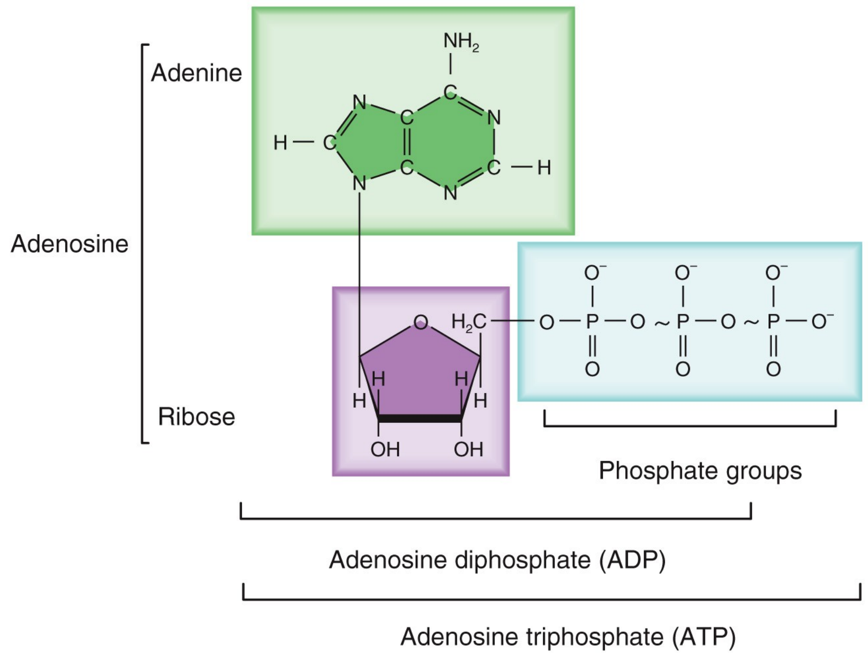
(b) Portion of a DNA molecule

DNA vs. RNA

Feature	DNA	RNA
Nitrogenous bases	Adenine (A), cytosine (C), guanine (G), **thymine (T)**.	Adenine (A), cytosine (C), guanine (G), **uracil (U)**.
Sugar in nucleotides	**Deoxyribose.**	**Ribose.**
Number of strands	Two (**double-helix**, like a twisted ladder).	One.
Nitrogenous base pairing (number of hydrogen bonds)	**A with T** (2), G with C (3).	**A with U** (2), G with C (3).
How is it copied?	**Self-replicating.**	**Made by using DNA as a blueprint.**
Function	**Encodes information for making proteins.**	**Carries the genetic code and assists in making proteins.**
Types	**Nuclear, mitochondrial.**	**Messenger RNA (mRNA), transfer**

Adenosine Triphosphate (ATP)

- ATP is the principal energy-providing molecule in the body
- Since phosphates are negatively-charged, they repel one another
  - Results in ATP being an unstable molecule
    - This is why energy must be stored in a more stable form – carbohydrates and fats



How Does The Cell Produce ATP?

- Cellular respiration is when glucose is catabolized
- Releases energy to remove phosphate to form ADP
- Two phases
  - Anaerobic (without oxygen); glucose partially broken down into pyruvic acid; yields 2 ATP
  - Aerobic (requires oxygen); glucose broken into CO_2 and H_2O); yields 30-32 ATP
  - Chapters 10 and 25 cover respiration

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